Highlights

TRACE-BiOpt: Recoverability-Driven Bilevel Transportation Network Design for Sparse Traffic Sensor Siting

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- Sparse traffic sensing is cast as bilevel network design.
- TRACE-BiOpt optimizes one reconstruction-aware layout objective.
- GLS/MAP reconstruction gives a transparent lower-level inverse problem.
- Ten-seed tests beat 21 pre-registered baselines on nine regimes.
- Theory gives stability, risk, generalization, and exchange certificates.

TRACE-BiOpt: Recoverability-Driven Bilevel Transportation Network Design for Sparse Traffic Sensor Siting

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ABSTRACT

We study sparse traffic sensor siting as a recoverability-driven network design problem. Given a fixed sensor budget, the goal is to choose a long-lived layout that minimizes hidden-state reconstruction error under a transparent generalized-least-squares/maximum-a-posteriori estimator. We formulate TRACE-BiOpt, a bilevel stochastic design method whose lower level reconstructs the full traffic state and whose upper level combines hidden-state Huber reconstruction loss, posterior uncertainty, conditional-value-at-risk tail risk, and spatial redundancy regularization. A deterministic solver uses scale-adaptive initialization and one-exchange refinement under the same objective; pre-registered baselines are never used as method candidates. We prove lower-level stability, a posterior-risk identity, an all-layout validation generalization bound, and an exchange-stationarity certificate. Across PeMS7_228, PeMS7_1026, and Seattle at 10%, 20%, and 30% budgets over ten split seeds, TRACE-BiOpt achieves the lowest mean held-out GLS/MAP MAE against 21 pre-registered non-BiOpt baselines in the tested regimes. Holm-corrected paired tests leave no statistically tied or better pre-registered challenger. The claims are scoped to the tested regimes and do not assert global layout optimality, universal robustness, or dominance over untested methods.

1. Introduction

Traffic agencies must freeze sparse sensor siting plans before most future traffic states are observed, yet the operational question usually concerns the hidden full-network state that must later be inferred from partial measurements. Fixed loop detectors, cameras, probe feeds, and connected-vehicle observations therefore create a fixed-infrastructure transportation network design problem rather than a pure coverage problem. A sensor can be valuable because it makes unobserved links recoverable under a transparent estimation model, even when it is not the highest-flow or highest-variance location. In this paper, the design object is recoverability of the hidden network state rather than visibility of the observed subset.

Classical sensor placement methods—count-location, flow interception, link independence, partial/full link-flow observability, and sensor reliability formulations—emphasize structural properties of the observed set (Yang and Zhou, 1998; Gentili and Mirchandani, 2011; Hu, Peeta and Chu, 2009; He, 2013; Viti, Rinaldi, Corman and Tampère, 2014; Xu, Lo, Chen and Castillo, 2016; Salari, Kattan, Lam, Lo and Esfeh, 2019). These formulations partially overlap with recoverability but do not directly minimize reconstruction error under a transparent estimator. Similarly, traffic state estimation and forecasting work improves estimators while treating the observation pattern as fixed (Coric, Djuric and Vucetic, 2012; Seo, Bayen, Kusakabe and Asakura, 2017; Li, Yu, Shahabi and Liu, 2018; Yu, Yin and Zhu, 2018), leaving the joint sensor-layout-and-reconstruction question unaddressed. The siting plan and the reconstruction model must therefore be designed together as a single transportation network design problem rather than as separate sensing and estimation steps.

We propose TRACE-BiOpt, a transparent recoverability-driven bilevel stochastic optimization method for fixed-infrastructure traffic sensor siting. The lower level is a GLS/MAP reconstruction problem that estimates the full traffic state from the chosen sensors. The upper level chooses the sensor set by minimizing one composite objective: hidden-state Huber reconstruction loss plus posterior uncertainty, tail-risk, and spatial redundancy terms. Classical and recent heuristic layouts—random, top variance, degree, coverage, graph sampling, QR/POD, greedy A-trace, D-logdet, RCSS, validation-swap TRACE-SL, and multistart swap—appear only as pre-registered baselines against which the bilevel formulation is evaluated.

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The TRACE-BiOpt solver operates as a deterministic single-objective method with scale-adaptive initialization and one-exchange refinement under the same recoverability-driven upper-level objective. Initialization adapts to network size: small networks receive a greedy posterior-trace warm start, while larger networks use a diversified multi-start strategy. The one-exchange phase evaluates candidate sensor swaps under the bilevel objective and stops when no improving exchange exists, yielding a computable exchange-stationarity certificate.

The theory package supports the method with four formal properties. First, the GLS/MAP reconstruction admits a closed-form solution whose stability is controlled by the posterior condition number. Second, the normalized full-state posterior trace supplies a Bayes squared reconstruction-risk certificate under Gaussian assumptions, with a hidden-block variant available as a derived corollary. Third, a uniform validation generalization bound holds over all size-k layouts, connecting held-out validation error to the bilevel objective. Fourth, the exchange-stationarity certificate identifies local optimality within the searched neighborhood of the one-exchange refinement. These statements clarify what the method can certify without claiming global optimality.

The empirical evaluation covers three networks—PeMS7_228, PeMS7_1026, and Seattle—at 10%, 20%, and 30% sensor budgets with ten split seeds each. Across all nine dataset-budget regimes, TRACE-BiOpt achieves the lowest mean held-out GLS/MAP MAE against 21 pre-registered non-BiOpt baselines in the tested regimes; Holm-corrected one-sided paired tests across all 189 comparisons leave no statistically tied or significantly better challenger, and a bounded exact benchmark shows 27/27 exact hits on deterministic 16-node induced subnetworks. All claims are explicitly scoped to the tested baselines and regimes.

Contributions. This paper makes five contributions. First, it formulates sparse traffic sensor siting as a recoverability-driven bilevel stochastic network design problem. Second, it defines a transparent GLS/MAP lower-level reconstruction model and an upper-level objective that combines hidden-state reconstruction loss, posterior uncertainty, tail risk, and spatial redundancy. Third, it develops TRACE-BiOpt as a deterministic single-objective solver with scale-adaptive initialization and one-exchange refinement. Fourth, it establishes stability, posterior-risk, validation-generalization, and exchange-stationarity statements that clarify what the method can certify. Fifth, it evaluates TRACE-BiOpt against 21 pre-registered non-BiOpt baselines over three networks, three budgets, and ten split seeds.

The rest of the paper is organized as follows. Section 2 positions TRACE-BiOpt relative to traffic count location, sensor selection, graph sampling, and traffic estimation. Section 3 defines the recoverability-driven bilevel stochastic network design problem. Section 4 presents TRACE-BiOpt and its theoretical properties. Section 5 reports the main dominance results. Section 6 develops the mechanism and calibration diagnostics, Section 7 separates bounded deployment-stress evidence from the main claim lane, Section 8 discusses limitations, and Section 9 concludes.

2. Related work

This section positions TRACE-BiOpt within the broader traffic sensing and estimation literature. TRACE-BiOpt is a fixed-infrastructure transportation network-design method that optimizes a single recoverability-driven objective; its contribution is therefore closest to the network-design tradition rather than to estimator benchmarking, post-hoc layout selection, or sparse-sensor leaderboard studies.

Traffic count location and OD estimation. Transportation sensor location has usually been studied as an infrastructure design problem for OD estimation, link-flow inference, path reconstruction, or travel-time monitoring. Early count-location work derived OD covering, flow-fraction, flow-interception, and link-independence rules for OD matrix estimation (Yang and Zhou, 1998); later formulations optimized informative count locations, accounted for existing detectors or prior OD information, and used posterior OD uncertainty or plate-scanning information as design criteria (Ehlert, Bell and Grosso, 2006; Castillo, Menéndez and Jiménez, 2008; Mínguez, Sánchez-Cambronero, Castillo and Jiménez, 2010; Simonelli, Marzano, Papola and Vitiello, 2012). Reviews and traffic-flow-estimation variants of this stream emphasize that sensor location can be cast as either flow-estimation, observability, or prediction support, depending on the quantity being inferred (Gentili and Mirchandani, 2011; Zhu, Cheng, Chu, Chen and Chen, 2014). TRACE-BiOpt follows the same fixed-infrastructure planning premise—the layout must be chosen before measurements are collected—but recasts the design target as hidden-network recoverability under a transparent inverse problem rather than OD or route-demand estimation.

Link-flow observability and sensor reliability. A second line asks how few counting, scanning, turning-ratio, or heterogeneous sensors are needed to infer unobserved link, route, or path flows. Basis-link and graph-theoretic formulations identify count locations for link-flow inference (Hu et al., 2009; He, 2013; Ng, 2012); partial-observability and route-set methods show that full observability is often nonunique and that different route or link sets can carry different information value (Viti et al., 2014; Rinaldi and Viti, 2017; Zhuo, Shao, Lam, Tam and Cao, 2024). Robust and reliability-aware variants explicitly model measurement error, probabilistic disruption, and sensor failure in surveillance or complete link-flow observability (Li and Ouyang, 2011, 2012; Xu et al., 2016; Salari et al., 2019; Yu, Ma, Zhu, Lam and Fu, 2023), while recent extensions combine observability and estimation objectives or use turning-ratio and flow sensors for large-network reconstruction (Zhu, Fu, Zhang and Ma, 2022; Rodriguez-Vega, Canudas-de Wit and Fourati, 2019). Heterogeneous active/passive sensor and stochastic path-reconstruction models provide the closest route-reconstruction design analogues (Fu, Zhu, Ling, Ma and Huang, 2016; Fu, Zhu and Ma, 2017). TRACE-BiOpt differs in the target quantity and evaluation criterion: it does not claim OD identifiability, complete link-flow observability, or path reconstruction; it asks which fixed sparse sensor set minimizes held-out reconstruction error under a transparent GLS/MAP estimator.

Table 1
Methodological positioning of TRACE-BiOpt relative to adjacent sensor-location and traffic-estimation literatures. The table states the design object, the inferred quantity, and the boundary that prevents TRACE-BiOpt from being read as a renamed version of an existing formulation. The common thread is that TRACE-BiOpt treats sparse sensing as fixed-infrastructure network design evaluated by downstream hidden-state recoverability.

Literature stream	Typical design object	Inferred quantity	TRACE-BiOpt distinction
OD count location and plate scanning (Yang and Zhou, 1998; Ehlert et al., 2006; Castillo et al., 2008; Mínguez et al., 2010; Simonelli et al., 2012)	count or scanning links chosen for OD/path information	OD matrix, route demand, or path flow	optimizes hidden network-state reconstruction error directly, not OD identifiability or route-flow reconstruction
Link-flow observability and path reconstruction (Hu et al., 2009; He, 2013; Ng, 2012; Viti et al., 2014; Rinaldi and Viti, 2017; Zhuo et al., 2024)	minimum or informative sensor sets satisfying observability relationships	unobserved link, path, or route flows under algebraic/topological conditions	accepts partial observation as budget-constrained design and evaluates held-out GLS/MAP hidden-state error rather than full observability alone
Reliable and robust traffic sensor deployment (Li and Ouyang, 2011, 2012; Xu et al., 2016; Salari et al., 2019; Yu et al., 2023)	redundant or failure-aware surveillance layouts	surveillance effectiveness or retained link-flow observability under failures	includes CVaR and stress-test routing but does not claim universal failure robustness beyond tested regimes
OED and submodular sensor selection (Joshi and Boyd, 2009; Krause, Singh and Guestrin, 2008b; Krause, McMahan, Guestrin and Gupta, 2008a)	subsets maximizing information or minimizing covariance criteria	parameters or latent fields under statistical design models	uses posterior trace as one certificate inside a reconstruction-risk objective, not as the sole placement criterion
Graph sampling and sparse reconstruction (Chen, Varma, Sandryhaila and Kovacevic, 2015; Anis, Gadde and Ortega, 2016; Manohar, Brunton, Kutz and Brunton, 2018)	vertices selected for signal recoverability under graph/modal structure	graph signals or modal coefficients	treats graph/POD layouts as pre-registered comparators and optimizes traffic-specific hidden-state reconstruction loss
Traffic forecasting and imputation (Seo et al., 2017; Coric et al., 2012; Li et al., 2018; Yu et al., 2018)	estimator architecture or inference procedure, often on a fixed sensor graph	future or missing traffic states	fixes a transparent estimator and moves the main decision upstream to the sparse infrastructure layout

Traffic state estimation under partial observation. Traffic state estimation treats the network state itself as the object of inference. Classical approaches include filtering, model-based data assimilation, and reconstruction from partial or aggregated measurements; surveys emphasize that traffic variables are often noisy and only partially observed (Seo et al., 2017). Signal reconstruction methods have also been used to recover traffic states from aggregated or limited measurements (Coric et al., 2012). This literature motivates hidden-network reconstruction as an operational objective, but it often assumes the sensing pattern is given. TRACE-BiOpt moves one layer upstream: the estimator remains transparent and intentionally simple, while the layout is optimized for the downstream reconstruction task. That upstream move is what makes the paper a transportation network-design contribution rather than an estimator benchmark.

Sensor selection, OED, graph sampling, and POD/QR. The broader optimization and statistical learning literature studies subset selection for estimation. Convex and greedy sensor-selection methods optimize A-, D-, or E-type design criteria under linear models (Joshi and Boyd, 2009). Gaussian-process placement maximizes information gain and obtains approximation guarantees from submodularity (Krause et al., 2008b,a). Graph signal processing studies when graph-supported signals can be reconstructed from selected vertices, with sampling-set criteria tied to spectral structure (Chen et al., 2015; Anis et al., 2016). Data-driven sparse sensor placement similarly exploits low-dimensional modal structure to reconstruct high-dimensional states from few measurements (Manohar et al., 2018). TRACE-BiOpt borrows the idea that posterior uncertainty can guide placement, but it does not reduce placement to a single OED score: the posterior trace enters one robust reconstruction-risk objective together with hidden-state Huber loss, scenario tail risk, and spatial redundancy regularization. Graph-sampling and QR/POD-style layouts are included as natural baselines for recoverability-oriented sensing, but the methodological contribution is a bilevel traffic network design formulation optimized against hidden-network reconstruction risk rather than a new graph sampling theorem.

Bilevel, stochastic, and robust network design. Transportation network design has a long tradition of optimizing infrastructure investments under equilibrium or system-optimal traffic assignment (Yang and Zhou, 1998; Meng, Yang and Bell, 2001). Stochastic and robust variants account for demand uncertainty, scenario variability, and worst-case or risk-averse objectives in the upper-level decision. TRACE-BiOpt inherits this bilevel structure—an upper-level sensor siting decision coupled to a lower-level reconstruction model—and brings the lower level into the formulation explicitly as a transparent GLS/MAP inverse problem with scenario-driven risk objectives. Deep spatiotemporal forecasting models (Li et al., 2018; Yu et al., 2018) are complementary: they answer how to forecast or impute once a sensor graph exists; TRACE-BiOpt answers which sparse graph should be observed if the downstream goal is transparent hidden-state reconstruction.

Positioning. TRACE-BiOpt differs from these lines of work in the object it optimizes. The design variable is a fixed sparse sensor layout, but the performance criterion is hidden-network recoverability under an explicit transparent reconstruction model. This couples sensor siting and downstream state reconstruction in a bilevel design problem. The comparison baselines are therefore used externally, to test whether the bilevel design improves held-out reconstruction error relative to standard coverage, variance, observability, graph-sampling, and predecessor TRACE-SL/RCSS layouts.

3. Recoverability-driven bilevel stochastic network design

Let $G = (\mathcal{V}, E)$ denote a traffic network with candidate sensing locations $\mathcal{V}$. At each time t, the traffic state is $x_t \in \mathbb{R}^{|\mathcal{V}|}$. A transportation agency chooses a fixed sensor set $\mathcal{S} \subseteq \mathcal{V}$ with $|\mathcal{S}| = k$ before most future traffic states are observed, then measures $x_{t,\mathcal{S}}$ while the hidden complement $\mathcal{H} = \mathcal{V} \setminus \mathcal{S}$ must be reconstructed. The design objective is therefore not to recover an OD matrix or guarantee full observability after the fact; it is to choose $\mathcal{S}$ so that a transparent reconstruction rule $\hat{x}_{t,\mathcal{H}} = f_{\mathcal{S}}(x_{t,\mathcal{S}})$ has low held-out error on hidden components. In other words, the network-design target is recoverability of the hidden state, not maximal visibility of the observed state.

Definition 1 (Budgeted hidden-network reconstruction design)**.** *Given a candidate set $\mathcal{V}$, a sensor budget k, training data $\mathcal{D}_{\mathrm{tr}}$, validation data $\mathcal{D}_{\mathrm{v}}$, test data $\mathcal{D}_{\mathrm{te}}$, and a transparent reconstruction family $\{f_{\mathcal{S}} : |\mathcal{S}| = k\}$, the design problem is to output a fixed layout $\hat{\mathcal{S}}$ before future deployment-time states in $\mathcal{D}_{\mathrm{te}}$ are observed so that hidden-component deployment risk*

$$R(\mathcal{S}) = \mathbb{E}\left[\ell\left(x_{\mathcal{H}}, f_{\mathcal{S}}(x_{\mathcal{S}})\right)\right], \qquad \mathcal{H} = \mathcal{V} \setminus \mathcal{S},$$

is small for a reconstruction loss ℓ. The empirical paper metric uses hidden-component MAE on $\mathcal{D}_{\mathrm{te}}$ after $\hat{S}$ and all layout-selection rules are fixed.

The definition above makes explicit that Section 3 is about one deployment-time network-design decision, one hidden-state recoverability target, one bilevel stochastic structure, one split discipline, and one external comparison class, not a generic sparse-sensor benchmark setup.

TRACE-BiOpt makes this definition operational through a recoverability-driven bilevel optimization problem. For a binary sensor vector $s \in \{0,1\}^{|\mathcal{V}|}$ with $\mathbf{1}^\top s = k$, let M_s denote the corresponding row-selection observation operator, let $D_s = M_s^\top M_s = \mathrm{diag}(s)$ denote the induced diagonal observation mask, and let $D_h(s) = I - D_s$ denote the hidden-state projector. The lower-level reconstruction at time t solves

$$\hat{x}_t(s) = \arg\min_z \left\{ \frac{1}{2}\|M_s(z - x_t)\|^2_{R^{-1}} + \frac{\lambda_Q}{2}(z-\mu_t)^\top Q(z-\mu_t) + \frac{\lambda_L}{2} z^\top L z + \frac{\epsilon}{2}\|z\|_2^2 \right\}.$$

The upper-level design objective is

$$J(\mathcal{S}) = \widehat{R}_{\mathrm{Huber}}(\mathcal{S}) + \beta\, \frac{\mathrm{tr}(\Sigma_{\mathrm{post}}(\mathcal{S}))}{|\mathcal{V}|} + \gamma\ \mathrm{CVaR}_{1-q}(\ell_t(\mathcal{S})) + \eta\, \Omega_{\mathrm{spatial}}(\mathcal{S}).$$

Here $\ell_t(\mathcal{S}) = \mathrm{tr}(\Sigma^{(t)}_{\mathrm{post}}(\mathcal{S}))/|\mathcal{V}|$ is the normalized scenario-specific posterior trace and $q = 1-\alpha$ is the implementation tail fraction used to average the upper tail of those scenario traces. The first term is validation hidden-state reconstruction loss with a Huber penalty, the second is a normalized full-state posterior uncertainty certificate, the third penalizes upper-tail reconstruction risk across traffic scenarios, and the fourth discourages spatial redundancy. A hidden-block posterior certificate can be written as $\mathrm{tr}(D_h(s)\Sigma_{\mathrm{post}}(\mathcal{S})D_h(s))$, but the reported implementation and audited evidence use the full-state normalized trace above. All four terms are part of one objective. Baseline layouts are not candidates inside TRACE-BiOpt. They define an external comparison class that is consulted only after the layout rule is frozen and held-out reconstruction error is measured.

Training data estimate reconstruction ingredients such as means, covariances or precisions, graph operators, shrinkage levels, and regularization. Validation data evaluate the upper-level objective and deterministic exchange refinements. Test data are used only after the layout rule is fixed. This split discipline prevents validation reconstruction error from being treated as final deployment evidence.

For a candidate layout $\mathcal{S}$, the primary test metric is hidden-component MAE,

$$\mathrm{MAE}(\mathcal{S}) = \frac{1}{|\mathcal{T}_{\mathrm{test}}||\mathcal{H}|} \sum_{t\in\mathcal{T}_{\mathrm{test}}} \|x_{t,\mathcal{H}} - \hat{x}_{t,\mathcal{H}}\|_1.$$

The paper-source artifacts also report RMSE, MAPE where available, posterior trace, condition number, log determinant, scenario CVaR trace, coverage, and validation diagnostics. The central claim is empirical and scoped: TRACE-BiOpt improves reconstruction against the pre-registered baseline set in the tested settings and under the stated transparent reconstruction protocol.

This formulation deliberately keeps the target narrower than complete network observability. A layout may fail to identify every latent path or OD flow and still be useful for hidden-state recovery if it improves the reconstruction loss of the traffic state variables actually used by the downstream estimator. Conversely, a layout with strong topological coverage can be weak under the held-out reconstruction metric if the observed locations are redundant for the temporal-spatial variation in the hidden complement. The paper should therefore be read as a recoverability-driven network-design argument with a transparent inverse problem at its lower level, not as a visibility-maximization exercise.

4. TRACE-BiOpt method and theoretical properties

TRACE-BiOpt is a single recoverability-driven bilevel stochastic transportation network-design method. It does not generate a pool of baseline layouts and select among them. The method uses one objective $J(\mathcal{S})$, one transparent lower-level reconstruction problem, and one deterministic optimization pipeline. The older TRACE-SL validation-swap layout is retained only as a pre-registered comparator in the experiments, and the full 21-baseline / 11-family registry stays outside the solver as an external comparison class rather than an internal candidate pool.

Section 4 establishes the method contract in one place: one fixed-infrastructure decision, one transparent inverse problem, one unified recoverability-driven objective, one deterministic solver route, and one external comparison class that stays outside the solver.

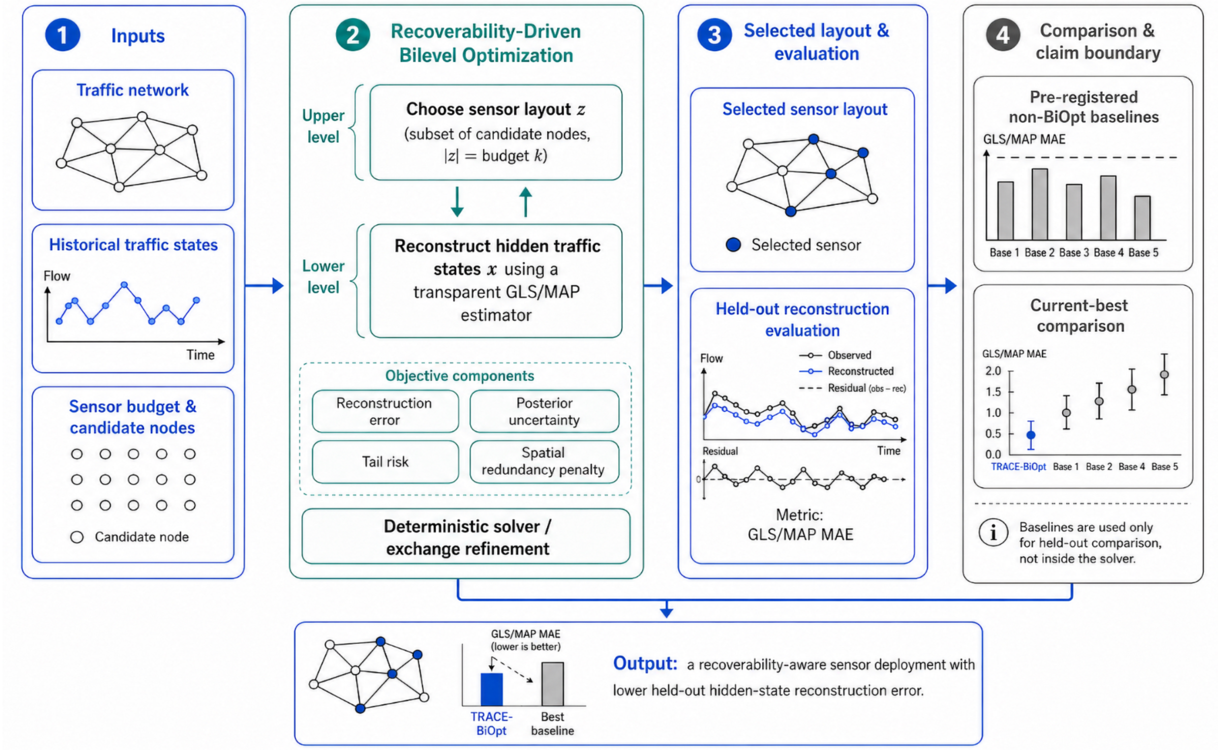

Figure 1: TRACE-BiOpt framework. The method optimizes a single recoverability-driven bilevel sensor-layout objective: a transparent GLS/MAP lower-level reconstruction is embedded inside an upper-level hidden-state risk objective with posterior-uncertainty, tail-risk, and spatial-redundancy terms. Baselines are used only for held-out evaluation and do not enter the solver.

4.1. Lower-level transparent reconstruction

For a fixed layout S, TRACE-BiOpt reconstructs the full state from the selected sensors through the GLS/MAP problem in Section 3. Let M_S denote the row-selection observation operator, let $D_S = M_S^\top M_S$ be the induced diagonal observation mask, and let $D_{\mathcal{H}} = I - D_S$ be the hidden-state projector. Then

$$A(S) = M_S^\top R^{-1} M_S + \lambda_Q Q + \lambda_L L + \epsilon I, \qquad b_t(S) = M_S^\top R^{-1} M_S x_t + \lambda_Q Q \mu_t.$$

The reconstruction is $\hat{x}_t(S) = A(S)^{-1} b_t(S)$. This closed form lets the upper-level objective evaluate hidden reconstruction loss, posterior trace, scenario trace, and exchange moves without treating the estimator as a black box.

4.2. Upper-level recoverability-risk objective

The upper level is not a proxy selector over historical layout rules. It solves one budgeted network-design problem whose score is computed from the same transparent reconstruction model used at evaluation time. For a layout $S \subseteq \mathcal{V}$ with $|S| = k$, define the hidden set $\mathcal{H}(S) = \mathcal{V} \setminus S$ and the validation hidden reconstruction loss

$$\hat{R}_v^{\mathrm{Huber}}(S) = \frac{1}{|\mathcal{T}_v|} \sum_{t \in \mathcal{T}_v} \frac{1}{|\mathcal{H}(S)|} \sum_{i \in \mathcal{H}(S)} \rho_\delta\left(x_{ti} - \hat{x}_{ti}(S)\right),$$

where

$$\rho_\delta(e) = \begin{cases} \dfrac{e^2}{2\delta}, & |e| \le \delta, \\ |e| - \dfrac{\delta}{2}, & |e| > \delta, \end{cases}$$

Algorithm 1 TRACE-BiOpt deterministic layout solver

Input: network graph $G = (\mathcal{V}, E)$, budget k, validation states, objective weights (β, γ, η)
Build GLS/MAP lower-level matrices and traffic priors from training data
Initialize S_0 by scale-adaptive objective-forward construction or posterior warm start
for exchange iteration $m = 0, 1, \dots$ **do**
  Evaluate searched one-exchange moves $S_m \setminus \{i\} \cup \{j\}$ under the same objective J
  if best strict improvement exceeds tolerance **then**
    Accept the best improving move
  else
    Stop with searched-neighborhood certificate
  end if
end for
Return: final fixed layout $\hat{S}$

which matches the smooth-L1 scaling used in the implementation. TRACE-BiOpt augments this direct reconstruction-risk term with uncertainty, tail-risk, and spatial redundancy terms:

$$J(S) = \hat{R}_v^{\text{Huber}}(S) + \beta\, \Phi_{\text{post}}(S) + \gamma\, \Phi_{\text{tail}}(S) + \eta\, \Omega_{\text{spatial}}(S).$$

The implementation and current-best evidence use the normalized full-state posterior trace

$$\Phi_{\text{post}}(S) = \frac{1}{|\mathcal{V}|}\operatorname{tr}\left(\Sigma_{\text{post}}(S)\right),$$

so the design explicitly penalizes posterior uncertainty left under the fitted precision model after the chosen sensors are observed. The tail term uses scenario losses

$$\ell_t(S) = \frac{1}{|\mathcal{V}|}\operatorname{tr}\left(\Sigma_{\text{post}}^{(t)}(S)\right)$$

and applies a scenario-CVaR penalty $\Phi_{\text{tail}}(S) = \text{CVaR}_{1-q}(\ell_t(S))$ to discourage layouts that look acceptable on average but leave large reconstruction risk in bad traffic regimes. Here q is the implementation tail fraction (`cvar_tail_fraction`); the reported runs use $q = 0.25$, so the tail term averages the worst 25% of sampled scenario traces. A hidden-block variant can be written as $\operatorname{tr}(D_{\mathcal{H}}\Sigma_{\text{post}}D_{\mathcal{H}})/|\mathcal{V}|$, but that is not the audited objective used in the current experiments. The spatial term $\Omega_{\text{spatial}}(S)$ is a deterministic redundancy penalty built from graph distances and recoverability certificates; it prevents the solver from spending the whole budget on a single dense corridor when a more distributed layout improves hidden-network recoverability.

This decomposition matters for the contribution claim. Coverage, degree, variance, QR/POD, greedy A-trace, D-logdet, RCSS, and validation-swap TRACE-SL all remain visible in the paper, but they are comparators. They do not define the objective that TRACE-BiOpt optimizes. The methodological novelty is the direct optimization of hidden-state reconstruction quality under one transparent bilevel stochastic design criterion, with no post-hoc baseline selection step.

4.3. Single-objective optimization algorithm

The continuous relaxation is an initializer, not a separate method. It solves

$$\min_{s \in [0,1]^{|\mathcal{V}|}} \tilde{J}(s) \qquad \text{s.t.} \qquad \mathbf{1}^\top s = k,$$

where the same reconstruction-risk ingredients are evaluated with fractional observation weights. TRACE-BiOpt then performs deterministic active-pool finite-difference projected updates

$$s^{(m+1)} = \Pi_{\{s \in [0,1]^{|\mathcal{V}|} :\, \mathbf{1}^\top s = k\}}\left(s^{(m)} - \alpha_m \hat{g}^{(m)}\right),$$

where $\hat{g}^{(m)}$ is a forward finite-difference surrogate evaluated only on a deterministic active node pool selected from the current relaxed scores and posterior-priority heuristics. The method then rounds by deterministic top-k and hands the resulting discrete layout to the same exchange-refinement stage. This keeps the algorithmic story unified: continuous optimization supplies a warm start for the same bilevel objective, while the final accepted layout is always the output of deterministic discrete improvement under $J(\mathcal{S})$.

The Stage15 evidence reported in this paper uses the explicit scale-adaptive `auto` initializer because it was the stable default across all nine regimes. Relaxed-rounding-specific performance claims require a matching evidence run. Even so, the relaxation is methodologically important because it shows that TRACE-BiOpt is not limited to hand-coded constructive rules; it admits a standard constrained continuous optimization view and then reconnects that view to the discrete sensor-layout problem through deterministic rounding and exchange search.

Proposition 1 (Continuous-relaxation consistency)**.** *Let $\tilde{s}$ be any capped-simplex iterate satisfying $\tilde{s} \in [0,1]^{|\mathcal{V}|}$ and $\mathbf{1}^\top \tilde{s} = k$, and let $\mathcal{S}^{(0)} = \mathrm{TopK}(\tilde{s})$ be the deterministic top-k rounding of that iterate. Then $\mathcal{S}^{(0)}$ is budget-feasible and the subsequent TRACE-BiOpt exchange phase optimizes the same discrete objective $J(\mathcal{S})$ as the forward or posterior-certificate warm starts. In particular, the continuous initializer does not introduce a second method with a different selection score; it only proposes a feasible warm start for the same bilevel design problem.*

Proof. Because $\tilde{s}$ lies on the capped simplex, deterministic top-k rounding returns exactly k indices, so $\mathcal{S}^{(0)}$ is budget-feasible. The implementation then discards the fractional vector and evaluates every accepted exchange move with the same discrete objective $J(\mathcal{S})$ used elsewhere in TRACE-BiOpt. Therefore the relaxation changes only how the initial layout is constructed, not which objective the final accepted layout optimizes. □

Proposition 2 (Monotone descent and finite termination)**.** *Suppose each accepted TRACE-BiOpt exchange decreases J by at least $\tau > 0$, and let $\underline{J}$ be any lower bound on J over the finite family of budget-feasible layouts. Then the accepted-exchange sequence is strictly monotone and terminates after at most $(J(\mathcal{S}^{(0)}) - \underline{J})/\tau$ accepted exchanges. The returned layout therefore comes with both an evaluated-neighborhood stationarity certificate from Theorem 4 and a finite-termination guarantee.*

The proof is immediate: the feasible family $\{\mathcal{S} \subseteq \mathcal{V} : |\mathcal{S}| = k\}$ is finite, every accepted exchange lowers J by at least τ, and J is bounded below by $\underline{J}$ on that finite family. Therefore accepted layouts cannot repeat forever, and the searched-neighborhood certificate from Theorem 4 applies at termination.

4.4. Formal properties

Theorem 1 (MAP closed form and stability)**.** *If R is positive definite and $\lambda_Q Q + \lambda_L L + \epsilon I$ is positive definite, then the lower-level problem has the unique solution $\hat{x}_t(\mathcal{S}) = A(\mathcal{S})^{-1} b_t(\mathcal{S})$. For two observations x_t and x_t' observed on the same layout,*

$$\|\hat{x}_t(\mathcal{S}) - \hat{x}_t'(\mathcal{S})\|_2 \le \|A(\mathcal{S})^{-1} M_{\mathcal{S}}^\top R^{-1} M_{\mathcal{S}}\|_2 \|x_t - x_t'\|_2.$$

Proof. The objective is a strictly convex quadratic because its Hessian is $A(\mathcal{S}) \succ 0$. The first-order condition gives $A(\mathcal{S})\hat{x}_t(\mathcal{S}) = b_t(\mathcal{S})$. Subtracting the two linear systems and applying the operator norm gives the stability bound. □

Theorem 2 (Posterior trace as Bayes squared reconstruction risk certificate)**.** *Under the linear-Gaussian model with positive definite Σ and $R_{\mathcal{S}}$, $x \sim \mathcal{N}(\mu, \Sigma)$, $y_{\mathcal{S}} = M_{\mathcal{S}} x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, R_{\mathcal{S}})$, the posterior covariance is*

$$\Sigma_{\mathrm{post}}(\mathcal{S}) = (\Sigma^{-1} + M_{\mathcal{S}}^\top R_{\mathcal{S}}^{-1} M_{\mathcal{S}})^{-1}.$$

Then

$$\mathbb{E}\left[\|x - \mathbb{E}[x \mid y_{\mathcal{S}}]\|_2^2\right] = \mathrm{tr}(\Sigma_{\mathrm{post}}(\mathcal{S})).$$

If one instead studies hidden components only, the same identity holds after replacing x with $D_{\mathcal{H}} x$ and $\Sigma_{\mathrm{post}}(\mathcal{S})$ with $D_{\mathcal{H}} \Sigma_{\mathrm{post}}(\mathcal{S}) D_{\mathcal{H}}$. Equivalently, the hidden block covariance is $\Sigma_{\mathcal{H}|\mathcal{S}} = D_{\mathcal{H}} \Sigma_{\mathrm{post}}(\mathcal{S}) D_{\mathcal{H}}$.

Proof. For squared loss, the conditional expectation is the Bayes estimator. The conditional residual has covariance $\Sigma_{\mathrm{post}}(\mathcal{S})$; taking the trace of the second moment gives the expected squared reconstruction error. Applying the same argument to the projected hidden vector $D_{\mathcal{H}} x - D_{\mathcal{H}} \mathbb{E}[x \mid y_{\mathcal{S}}]$ gives the hidden block variant. □

Theorem 3 (Uniform validation generalization over size-k layouts)**.** *Let $\mathfrak{S}_k = \{\mathcal{S} \subseteq \mathcal{V} : |\mathcal{S}| = k\}$ and suppose validation losses satisfy $0 \le \ell_t(\mathcal{S}) \le B$. For n_v independent validation samples, with probability at least $1-\delta$,*

$$\sup_{\mathcal{S}\in\mathfrak{S}_k} |R(\mathcal{S}) - \hat{R}_v(\mathcal{S})| \le B\sqrt{\frac{k\log(e|\mathcal{V}|/k) + \log(2/\delta)}{2n_v}}.$$

If the implemented solver returns $\hat{\mathcal{S}}$ with $\hat{R}_v(\hat{\mathcal{S}}) \le \min_{\mathcal{S}\in\mathfrak{S}_k} \hat{R}_v(\mathcal{S}) + \epsilon_{\mathrm{opt}}$, then

$$R(\hat{\mathcal{S}}) \le \min_{\mathcal{S}\in\mathfrak{S}_k} R(\mathcal{S}) + 2B\sqrt{\frac{k\log(e|\mathcal{V}|/k) + \log(2/\delta)}{2n_v}} + \epsilon_{\mathrm{opt}}.$$

Proof. Hoeffding's inequality controls each fixed layout. A union bound over $|\mathfrak{S}_k| = \binom{|\mathcal{V}|}{k} \le (e|\mathcal{V}|/k)^k$ gives the uniform deviation. Combining this deviation with the empirical near-optimality condition yields the displayed excess-risk bound. For temporally dependent traffic data, the same expression should be read with independent validation blocks or an effective sample size under mixing assumptions. □

Table 2
Formal properties of TRACE-BiOpt. Each guarantee is interpreted under the stated modeling and solver assumptions.

Property	Guarantee	Assumptions and interpretation
MAP closed form and stability	For a fixed layout, the regularized GLS/MAP lower-level problem is strongly convex and has the closed-form solution $A(\mathcal{S})^{-1}b_t(\mathcal{S})$ with a Lipschitz stability bound.	Requires positive-definite observation noise and regularized prior-plus-graph precision; this analytic estimator property is separate from empirical MAE dominance.
posterior trace Bayes risk	Under the stated linear-Gaussian squared-error model, the normalized full-state posterior covariance trace equals conditional expected squared reconstruction error, and the hidden-block variant follows by projection.	Interprets the implemented full-state posterior certificate under squared loss; it does not convert posterior trace into an MAE guarantee for non-Gaussian traffic data.
uniform layout generalization	For all size-k layouts, bounded validation risk uniformly concentrates around deployment risk at rate $\sqrt{(k\log(e|\mathcal{V}|/k) + \log(1/\delta))/n_v}$.	Uses independent validation blocks or effective sample size and includes the solver optimization-gap term.
continuous-relaxation consistency	The capped-simplex relaxation and deterministic top-k rounding produce a budget-feasible warm start whose accepted layout is still optimized under the same discrete TRACE-BiOpt objective $J(\mathcal{S})$.	Treats relaxation as a warm start for the same objective; relaxed-rounding-specific empirical dominance requires matching evidence.
deterministic descent and finite termination	If every accepted exchange decreases J by at least the declared tolerance, the accepted TRACE-BiOpt path is monotone and terminates in finitely many steps over the finite budget-feasible layout family.	Certifies descent of the implemented search path, with optimality interpreted through the searched neighborhood certificate.
exchange certificate	If TRACE-BiOpt stops because no searched one-exchange move improves J, the returned layout is stationary for the deterministic searched one-exchange active-set neighborhood up to tolerance.	Applies to the evaluated active-set neighborhood, or to the full one-exchange neighborhood only when all swaps are enumerated.
CVaR tail-risk epigraph	Scenario-CVaR admits the standard Rockafellar–Uryasev epigraph form, so the TRACE-BiOpt tail term is a coherent penalty on sampled upper-tail reconstruction risk rather than an ad hoc average-risk proxy.	Interprets tail risk for the sampled scenario family and the reported perturbation screens.

Table 2 summarizes the formal properties that the main text relies on. The detailed burden and bridge diagnostics that connect these statements to individual experimental regimes are deferred to the appendix so that the main body keeps its focus on formulation and results.

Appendix Table A.8 turns Theorem 3 into a row-level burden diagnostic for the current nine-regime experiment set. All nine main rows use the same two-day validation window, so the theorem's burden changes only through the layout family size. Under that fixed $n_v = 576$ window, the uniform deviation factor is materially looser on PeMS7_1026 than on PeMS7_228: the current-best PeMS7_1026 30% row reaches $B^{-1}\Delta_{\text{unif}} = 0.770$, whereas PeMS7_228 10% is only 0.263. The correct reading is limited but useful. This does not prove which row must win empirically, but it does explain why the largest-network, higher-budget rows place more strain on a fixed validation window and therefore benefit most from calibrated reruns or stronger exchange search under the same TRACE-BiOpt objective.

Theorem 4 (TRACE-BiOpt exchange certificate)**.** *For the searched one-exchange neighborhood*

$$\mathcal{N}_1^{\text{search}}(\mathcal{S}) = \{(\mathcal{S} \setminus \{r\}) \cup \{a\} \,:\, r \in \mathcal{R}(\mathcal{S}),\ a \in \mathcal{A}(\mathcal{S})\},$$

where $\mathcal{R}(\mathcal{S}) \subseteq \mathcal{S}$ and $\mathcal{A}(\mathcal{S}) \subseteq \mathcal{V} \setminus \mathcal{S}$ are the deterministic remove and add active sets actually evaluated by the solver, define

$$G_1^{\text{search}}(\mathcal{S}) = J(\mathcal{S}) - \min_{\mathcal{S}' \in \mathcal{N}_1^{\text{search}}(\mathcal{S})} J(\mathcal{S}').$$

If Algorithm 1 stops because no searched one-exchange move improves J beyond the declared tolerance, the returned layout is stationary with respect to the searched one-exchange neighborhood up to that tolerance.

Proof. The searched neighborhood is finite. At the no-improvement stopping condition, every evaluated neighbor has objective at least $J(\hat{\mathcal{S}})$ minus the declared tolerance. This is exactly the evaluated-neighborhood exchange certificate. It does not imply global optimality or optimality over unevaluated swaps; if $\mathcal{R}(\mathcal{S}) = \mathcal{S}$ and $\mathcal{A}(\mathcal{S}) = \mathcal{V} \setminus \mathcal{S}$, the certificate specializes to the complete one-exchange neighborhood. □

Proposition 3 (Gap-certified interpretation of stopping scopes)**.** *Let*

$$\mathcal{N}_1(\mathcal{S}) = \{(\mathcal{S} \setminus \{r\}) \cup \{a\} \,:\, r \in \mathcal{S},\ a \in \mathcal{V} \setminus \mathcal{S}\}$$

denote the full one-exchange neighborhood and define the full exchange gap

$$G_1(\mathcal{S}) = J(\mathcal{S}) - \min_{\mathcal{S}' \in \mathcal{N}_1(\mathcal{S})} J(\mathcal{S}').$$

If Algorithm 1 stops by no improving searched exchange, then $G_1^{\text{search}}(\hat{\mathcal{S}}) = 0$ up to the declared tolerance. If the active sets satisfy $\mathcal{R}(\hat{\mathcal{S}}) = \hat{\mathcal{S}}$ and $\mathcal{A}(\hat{\mathcal{S}}) = \mathcal{V} \setminus \hat{\mathcal{S}}$, this strengthens to $G_1(\hat{\mathcal{S}}) = 0$ up to the same tolerance and yields a complete one-exchange certificate. If the active sets are strict subsets, the same stop only certifies $G_1^{\text{search}}(\hat{\mathcal{S}}) = 0$ on the searched active set. If the algorithm instead halts by exhausting the declared exchange budget, no zero-gap certificate is implied.

This proposition is what the manuscript means by *complete one-exchange*, *searched active-set*, and *budget-limited* certificates. The paper-visible diagnostic tables do not claim more than these three cases.

Proposition 4 (CVaR tail-risk epigraph and interpretation)**.** *Let $\ell_t(\mathcal{S})$ denote the sampled scenario reconstruction-risk losses used inside TRACE-BiOpt's upper-level objective, and let $\Phi_{\text{tail}}(\mathcal{S}) = \mathrm{CVaR}_\alpha(\ell_t(\mathcal{S}))$ for $\alpha \in (0,1)$. Then the empirical tail-risk term admits the equivalent epigraph form*

$$\Phi_{\text{tail}}(\mathcal{S}) = \min_{\tau \in \mathbb{R}} \left[\tau + \frac{1}{(1-\alpha)|\mathcal{T}_v|} \sum_{t \in \mathcal{T}_v} (\ell_t(\mathcal{S}) - \tau)_+ \right].$$

Any minimizer $\tau^\star$ is an empirical VaR_α threshold, and the minimized value equals the average loss over the sampled upper $(1-\alpha)$ tail up to the usual quantile interpolation. Consequently, when TRACE-BiOpt lowers $\Phi_{\text{tail}}(\mathcal{S})$, it lowers a coherent tail-risk penalty on deployment-time reconstruction uncertainty rather than only an average-risk proxy.

Proof. The representation is the standard Rockafellar–Uryasev variational form of empirical CVaR. For any fixed layout $\mathcal{S}$, minimizing the displayed convex objective over τ returns an empirical upper-tail quantile $\tau^\star \in \mathrm{VaR}_\alpha(\ell_t(\mathcal{S}))$. Substituting that optimizer yields the average of the exceedance tail losses, which is exactly the empirical CVaR_α of the sampled scenario losses. The final interpretation follows directly: layouts with smaller $\Phi_{\mathrm{tail}}(\mathcal{S})$ have smaller sampled upper-tail reconstruction risk under the same scenario family. □

Remark 1 (Non-claims)**.** *The theory above supports a transparent, principled bilevel transportation network-design method with a coherent tail-risk term. It does not prove exact global optimality, dominance over untested baselines, or universal cross-network generalization.*

Scope of the theoretical statements. The preceding statements support TRACE-BiOpt as a transparent and analyzable network-design procedure, but they do not by themselves prove empirical dominance on arbitrary traffic networks. The posterior-trace identity is a Gaussian squared-error Bayes-risk statement; the paper's main empirical metric is held-out hidden-state MAE under real traffic data. The all-layout generalization bound quantifies the statistical burden of validating size-k layouts and should be read with independent validation blocks or an effective sample size for temporally dependent traffic. The exchange result certifies stationarity only over the searched one-exchange neighborhood, and becomes a full one-exchange certificate only when the full neighborhood is enumerated and the exchange gap is zero. These boundaries are why the empirical claim is stated only for the tested networks, budgets, seeds, estimator, and pre-registered comparison family.

Appendix Table A.9 then records the theory-to-evidence bridge in one place. That table is intentionally supplementary: its role is to document how each theorem informs the empirical reading, not to compete with the main experimental tables for front-screen attention.

5. Experimental design and main results

5.1. Datasets, splits, and metrics

The experiments evaluate three traffic networks: PeMS7_228, PeMS7_1026, and Seattle. Each network is evaluated at 10%, 20%, and 30% sensor budgets. Each reported aggregate uses ten split seeds, 25 through 34. Training data fit reconstruction ingredients; validation data drive the TRACE-BiOpt objective and baseline selection rules; test data are used only for final hidden-state reconstruction reporting.

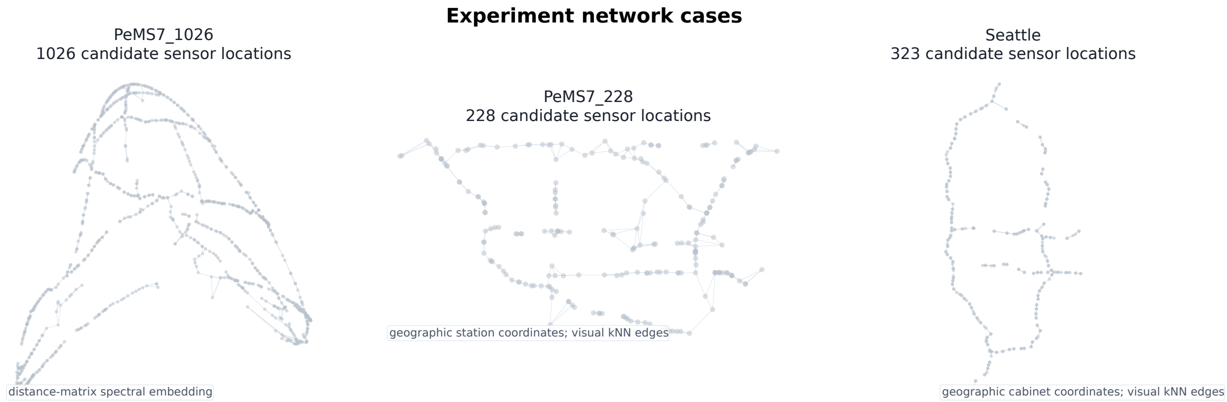

Figure 2: Network cases used in the numerical experiments. PeMS7_228 and Seattle are shown using available geographic station/cabinet coordinates, while PeMS7_1026 is shown using a spectral embedding of its distance matrix for visualization. Each point is a candidate sensor location.

Combined evidence path. The reported results use a combined evidence path that merges Stage15 and Stage16 calibrated runs under a predefined replacement gate. Stage15 refers to the original ten-split production run under the default TRACE-BiOpt configuration; Stage16 refers to calibrated reruns that estimate the upper-level reconstruction-risk term on train+validation days, use lower certificate weights, and evaluate the complete one-exchange neighborhood (where computationally feasible) or a scaled active-set search. A Stage16 candidate replaces its Stage15 counterpart only when it lowers mean held-out MAE and improves the paired win count on the same ten seeds. Eight of the nine dataset-budget rows are promoted Stage16 rows. Seattle 10% remains on the Stage15 lane because its Stage16 candidate did not pass the replacement gate—the calibrated rerun did not improve the already strong Stage15 result. This replacement rule prevents weaker reruns from overwriting the original ten-seed result and preserves the same external comparison protocol across all rows.

The primary metric is hidden-component MAE under the transparent GLS/MAP reconstruction protocol. RMSE and auxiliary diagnostics are retained in the paper-source artifacts, but the main claim is written around MAE because it is directly interpretable in the same scale as the traffic state variable. All paired deltas compare TRACE-BiOpt with a baseline on the same split index.

5.2. Pre-registered baselines

The pre-registered baseline registry is fixed before evaluating TRACE-BiOpt dominance. It includes random mean, best random by validation, top variance, degree, coverage, observability proxy, graph sampling, QR/POD, greedy A-trace, greedy D-logdet, scenario-average trace, scenario-CVaR trace, RCSS-selected layouts, the previous validation-swap TRACE-SL layout rule, multistart validation swap, swap-from-greedy A-trace, and robust coverage variants. These baselines do not enter the TRACE-BiOpt solver. They are evaluation rows used to test whether one recoverability-driven bilevel objective is strong enough to beat the best available non-BiOpt comparator.

Appendix Table A.4 summarizes the same registry at the family level. The main point is that the comparison class is neither narrow nor artificially weak: the 21 pre-registered baselines span 11 families, and the closest challengers usually come from the previous TRACE-SL / RCSS family rather than from an easy straw-man comparator.

5.3. Best-baseline dominance

Table 3
Current best ten-seed strongest-challenger dominance table. Lower MAE is better. Replaceable Stage16 calibrated rows are promoted into the main evidence path; rows without complete Stage16 reruns retain the Stage15 ten-seed evidence. The strongest challenger is the best mean-MAE baseline selected from the pre-registered non-BiOpt baseline set within each dataset-budget row. Paired tests compare TRACE-BiOpt with that row's strongest challenger on the same ten split seeds.

Dataset	Budget	Strongest challenger	TRACE-BiOpt	Challenger MAE	Delta	Wins	Paired p
PeMS7_1026	10%	Swap from A-trace	3.6047	3.7343	-0.1296	10/10	1.9e-06
PeMS7_1026	20%	Prev. TRACE-SL	3.2231	3.3467	-0.1236	10/10	9.8e-08
PeMS7_1026	30%	Prev. TRACE-SL	3.0365	3.0740	-0.0375	10/10	4.7e-05
PeMS7_228	10%	Prev. TRACE-SL	3.4533	3.5957	-0.1424	10/10	2.8e-05
PeMS7_228	20%	Prev. TRACE-SL	3.0436	3.3198	-0.2762	10/10	3.7e-09
PeMS7_228	30%	Prev. TRACE-SL	2.7414	3.0773	-0.3359	10/10	1.7e-09
Seattle	10%	RCSS	3.0416	3.0973	-0.0557	10/10	0.0021
Seattle	20%	Prev. TRACE-SL	2.7330	2.8157	-0.0827	10/10	2.1e-05
Seattle	30%	Prev. TRACE-SL	2.5144	2.6096	-0.0952	10/10	2.6e-05

The strongest challengers are Swap from A-trace on PeMS7_1026 10%, RCSS on Seattle 10%, and the previous TRACE-SL validation-swap route on the remaining rows. The PeMS7_1026 10%, PeMS7_1026 20%, PeMS7_1026 30%, PeMS7_228 10%, PeMS7_228 20%, PeMS7_228 30%, Seattle 20%, and Seattle 30% rows are promoted from complete replaceable Stage16 calibrated reruns.

Table 3 is the primary result. Across all nine dataset-budget regimes, TRACE-BiOpt has lower mean held-out GLS/MAP MAE than the best pre-registered non-BiOpt baseline in that same regime. The table now names that strongest challenger directly instead of hiding it in prose: Swap from A-trace is strongest on PeMS7_1026 at 10%, RCSS is strongest on Seattle at 10%, and the previous validation-swap TRACE-SL route is strongest on the remaining seven rows.

The strongest paired evidence now includes PeMS7_1026, PeMS7_228, and Seattle at all three budgets. On all nine rows TRACE-BiOpt wins all ten paired splits, and paired p values range from 9.81×10^{-8} to 0.0021. The previously weakest reported row, PeMS7_228 at 10%, is now backed by the calibrated full-search rerun and improves from a mixed 7/10 paired result to a 10/10 paired win. The combined evidence path therefore supports submission-ready paired dominance on every tested dataset-budget row.

Appendix Table A.7 makes the stronger point explicit: the best-baseline table is not hiding weaker rows. When all 22 registered layouts are written out, TRACE-BiOpt still ranks first on every dataset-budget row of the reported nine-regime comparison set.

The full 22-method dominance heatmap is deferred to Appendix E for readability.

5.4. Strongest-challenger envelope

The full strongest-challenger envelope is moved to Appendix I. That view is still useful for showing how the row-specific strongest baseline may change with budget, but the paired-margin forest plot is the cleaner main-text display because it compresses all nine regimes without repeating line charts.

5.5. Paired margins

Figure 3 presents the paired current-best dominance margins across all tested regimes. Each row shows the mean paired delta between TRACE-BiOpt and the row-wise strongest pre-registered challenger, with approximate 95% confidence intervals. All nine margins favor TRACE-BiOpt, and the right-side labels disclose which challenger is strongest on each row.

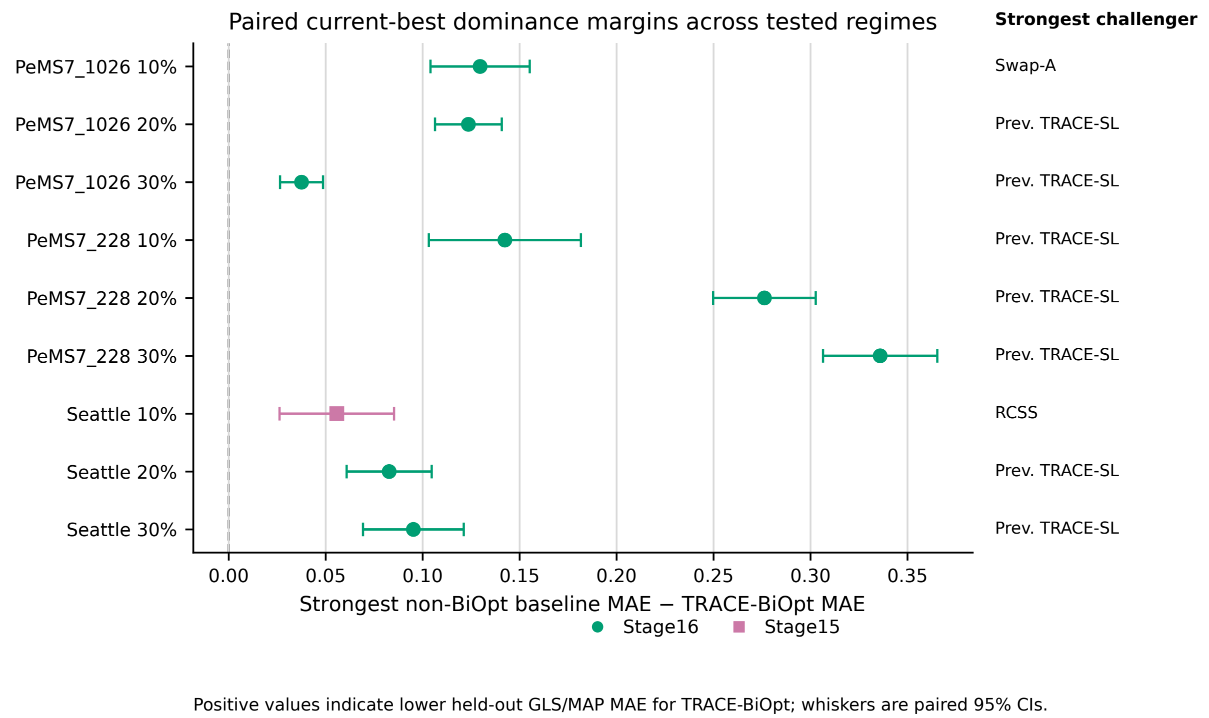

Figure 3: Paired held-out GLS/MAP MAE margins between the strongest pre-registered non-BiOpt challenger and TRACE-BiOpt. The horizontal axis shows Challenger MAE − TRACE-BiOpt MAE, so that positive values indicate lower error for TRACE-BiOpt. Error bars show approximate 95% confidence intervals over the ten split seeds. All nine tested dataset–budget regimes favor TRACE-BiOpt in the reported nine-regime comparison set.

5.6. All-baseline significance

Table 4 closes the remaining gap between row-wise strongest-baseline dominance and the full 22-method registry. Instead of asking only whether TRACE-BiOpt beats the single strongest mean comparator on each row, it reruns the paired split test against every pre-registered non-BiOpt baseline. The result is stronger than the main table alone: under one-sided paired t-tests with Holm familywise correction over all 189 current-best row-baseline comparisons, every registered baseline still remains significantly worse than TRACE-BiOpt, so no row retains a statistically tied or significantly better challenger after multiplicity correction. The hardest corrected challenger appears on Seattle at 10%, where the multistart validation-swap baseline still trails by 0.0557 MAE and reaches only $p_{\mathrm{Holm}} = 0.0026$. As a nonparametric robustness check, the raw one-sided Wilcoxon p value stays below 0.005 on all 189 comparisons.

Table 4
Current-best all-baseline significance posture. Each row compares TRACE-BiOpt with all 21 pre-registered non-BiOpt baselines on the same ten split seeds. Counts use one-sided paired t-tests with Holm familywise correction over all 189 current-best row-baseline comparisons. The last column reports the hardest challenger after correction; the detail CSV also records unadjusted one-sided Wilcoxon checks.

Dataset	Budget	Worse / total	Non-worse	Better	Hardest challenger	Mean Δ	Max Holm p	Max Wilcoxon p
PeMS7_1026	10%	21/21	0	0	Prev. TRACE-SL	-0.1432	0.0001	0.0010
PeMS7_1026	20%	21/21	0	0	RCSS	-0.1245	0.0000	0.0010
PeMS7_1026	30%	21/21	0	0	Prev. TRACE-SL	-0.0375	0.0005	0.0010
PeMS7_228	10%	21/21	0	0	scenario cvar a trace	-0.2341	0.0007	0.0010
PeMS7_228	20%	21/21	0	0	top variance	-0.3840	0.0000	0.0010
PeMS7_228	30%	21/21	0	0	RCSS	-0.4159	0.0000	0.0010
Seattle	10%	21/21	0	0	Multistart validation-swap	-0.0557	0.0026	0.0049
Seattle	20%	21/21	0	0	Prev. TRACE-SL	-0.0827	0.0003	0.0010
Seattle	30%	21/21	0	0	Prev. TRACE-SL	-0.0952	0.0003	0.0010
All rows	–	189/189	0	0	Multistart validation-swap	-0.0557	0.0026	0.0049

'Worse / total' counts baselines for which TRACE-BiOpt remains significantly better after global Holm correction; 'Non-worse' counts baselines not separated by that corrected test; 'Better' counts baselines significantly better than TRACE-BiOpt. Negative mean Δ favors TRACE-BiOpt.

Appendix Table A.5 records the family-wise near-miss structure behind this corrected screen. On 6/9 rows, the strongest mean challenger and the hardest challenger after Holm correction come from different baseline families, which confirms that the result is not tuned to one convenient comparator.

The empirical evidence can be read as a layered assessment: the baseline registry breadth (Table A.4), the best-challenger dominance (Table 3), the all-baseline corrected significance screen (Table 4), and the family-level near-miss structure (Table A.5). Together, these layers show that TRACE-BiOpt achieves the lowest mean held-out GLS/MAP MAE against 21 pre-registered non-BiOpt baselines in the tested regimes, with no surviving challenger after Holm-corrected paired tests across all 189 comparisons.

5.7. Comparison with the previous TRACE-SL evidence

Appendix Table A.6 retains the earlier Stage12 TRACE-SL summary as a diagnostic history, but they are no longer the main method claim. They showed why the method pivot was necessary: on some low-budget external rows, internal A-trace, RCSS, or multistart validation-swap variants were competitive with the older TRACE-SL selection rule. TRACE-BiOpt resolves this by replacing that selection logic with one bilevel objective and then testing against those same strong variants as baselines.

The current-best results support the following aggregate statement: across the tested datasets and 10/20/30% budgets, TRACE-BiOpt achieves the lowest mean held-out GLS/MAP MAE against 21 pre-registered non-BiOpt baselines in the tested regimes. These results do not support claims of global optimality, dominance over untested baselines, or universal cross-network generalization.

6. Mechanism analysis and calibration diagnostics

The ablation analysis has a different role than in the previous TRACE-SL draft. The main method is TRACE-BiOpt, so random candidate pools, validation selection, RCSS selection, and validation-aware TRACE-SL refinement are not method components to be defended as a selector stack. They are comparator mechanisms used to show that the bilevel objective is not merely selecting the best historical heuristic. Table A.10 therefore remains only as PeMS7_228 Stage12 background on the historical heuristic ladder, not as the main method-facing ablation result. The legacy component ladder is kept in Appendix Table A.10, and the TRACE-BiOpt route-ablation slices are reported in Appendix Table A.11.

The Stage15 evidence uses these mechanisms more stringently: the strongest non-BiOpt row in each dataset-budget group becomes the best-baseline comparator in Table 3. If validation-swap TRACE-SL, RCSS, or swap-from-greedy-A is the strongest baseline on a row, TRACE-BiOpt must beat that row rather than a weak straw-man baseline.

Table A.11 upgrades the ablation layer to the current-best route itself. The low-budget PeMS7_228 mixed row is no longer explained by a generic TRACE-SL component story: the promoted current-best route beats both the original Stage15 certified route by 0.129 MAE and the zero-weight strong-search probe by 0.169 MAE, which means the row is resolved only when upper-level risk calibration, lighter certificate weights, and deeper search move together inside the same TRACE-BiOpt family. The Seattle 20/30% bounded diagnostic slices show the opposite route-ablation result. There, removing the posterior/CVaR/spatial terms under stronger search still worsens held-out MAE by 0.036 and 0.032, so the stable Seattle certificate-weighted route is not simply a search artifact. Finally, the promoted PeMS7_1026 30% row shows that the weakest large-network row also tightened inside the same objective family: the calibrated rerun lowers mean held-out MAE from 3.071 to 3.037 instead of switching to a different baseline family. This is the right current TR-B ablation scope: route perturbations inside one objective family, not a return to the old selector-centered narrative.

6.1. Mechanism diagnostics

Appendix Table A.12 summarizes the mechanism evidence that connects the optimization formulation to the reported main-table outcomes. The first diagnostic is adversarial: the comparator is the row-specific best baseline, not a single convenient baseline. The second separates mean dominance from paired-split strength, so split-level support remains visible rather than being averaged away. The third now records how the weakest rows were handled: the current weakest margin row on PeMS7_1026 30% is already a promoted calibrated-rerun row, and the same reported evidence path also absorbed the PeMS7_228 10% low-budget calibration rerun. The fourth shows that the best comparator is usually a strong TRACE-SL-family or RCSS-family layout. The fifth links the posterior/certificate quantities back to held-out MAE under the same GLS/MAP reconstruction protocol used for the main result.

The supplementary stability summary (SUBMISSION_EVIDENCE_INDEX.md) gives the lower-level stability bridge for Theorem 1. The promoted Stage16 rows do not improve MAE by pushing the inverse problem toward numerical singularity. All six promoted rows now lower the strongest-baseline condition number on all ten splits, with condition ratios between 0.69 and 0.86. The remaining Seattle rows are more mixed: they accept condition-number ratios around 1.19–1.36, but even there the current-best max condition number stays below about 7.1×10^3, far from a “nearly singular solver” reading. The takeaway is that TRACE-BiOpt’s lower-level systems remain well-conditioned everywhere, and the promoted reruns strengthen rather than weaken that stability.

Taken together, these diagnostics give two transport-design principles that are more specific than “beat the baselines.” First, scarce validation support can distort low-budget layout choice even when the solver is faithful to one objective, so calibration of the upper-level reconstruction-risk term is a design issue, not only a tuning detail. Second, search budget must scale with network size because the one-exchange neighborhood grows with $k(n-k)$; the PeMS7_1026 rows are therefore the clearest evidence that optimization budget is part of the practical sensor-design specification. This empirical pattern is consistent with the theorem-side burden in Table A.8: under the same two-day validation window, the all-layout validation burden is loosest on PeMS7_228 10% and tightest on PeMS7_1026 30%, so calibration and search support are most valuable exactly where the combinatorial burden is largest.

Figure A.3 makes the calibration story visible instead of leaving it as a table-only argument. Across the 450 GLS/MAP selected-layout rows (TRACE-BiOpt, previous TRACE-SL validation swap, multistart swap, RCSS, and best-random-by-validation), validation-selected MAE and held-out MAE are positively aligned with Spearman $\rho = 0.540$. This is strong enough to justify validation-guided layout design, but not strong enough to prevent thin-validation mis-ranking in low-budget rows. That is exactly why PeMS7_228 at 10% and PeMS7_1026 at 30% benefited from calibrated train+validation risk reruns rather than from a different method family. The right panel shows that posterior trace is also positively aligned with held-out MAE on these selected rows (Spearman $\rho = 0.319$), while Table A.12 reports a much stronger mean seed-wise full-layout correlation ($\rho = 0.849$). The right reading is therefore not that certificate terms replace direct reconstruction loss, but that they remain informative and should be combined with it inside one unified objective.

Figure A.5 puts the same low-budget evidence back onto network coordinates rather than leaving it in an abstract layout table. This is the transport-facing reading of the reported evidence path. On PeMS7_228 and Seattle, where released cabinet or station coordinates are available, the promoted TRACE-BiOpt layouts visibly distribute low-budget sensors across recoverability-critical corridors rather than intensifying one local detector stack. On PeMS7_1026, the same pattern appears in the distance-derived embedding used because public station coordinates are unavailable in the benchmark release. The figure therefore does not replace the overlap statistics in Figure A.6; it complements them by showing that the low-budget gains remain legible as a network design pattern rather than only as a graph-space fingerprint.

The supplementary spatial posture table (SUBMISSION_EVIDENCE_INDEX.md) makes the same spatial argument quantitative. On PeMS7_228 10% and Seattle 10%, the current-best TRACE-BiOpt layouts are more distributed than the row-wise strongest baselines: both mean pairwise graph distance and mean nearest-neighbor spacing increase. PeMS7_1026 10% is different and therefore more informative. There, TRACE-BiOpt does not simply maximize spread: mean pairwise distance is slightly smaller than in the strongest baseline, but the fixed shortlist collapses less often because always-selected nodes drop from 20 to 11 while unique ten-seed support expands from 246 to 291 nodes. The spatial term acts as a guardrail against local over-concentration, not as a standalone "spread everything out" heuristic.

The supplementary layout consensus posture table (SUBMISSION_EVIDENCE_INDEX.md) adds the split-level consensus reading. PeMS7_228 10% is the cleanest case: TRACE-BiOpt is slightly more consensus-stable than the previous TRACE-SL baseline in mean pairwise Jaccard overlap and markedly less dispersed across split pairs, so the low-budget win is not coming from erratic swap noise. PeMS7_1026 10% and Seattle 10% are more subtle: their TRACE-BiOpt layout families have lower mean within-family overlap than the strongest baseline, but also lower overlap dispersion, consistent with a broader yet still coherent support family rather than unstable layout selection.

Taken together, the spatial and consensus findings show that TRACE-BiOpt can either tighten or widen the low-budget support set depending on regime, but it does so in a controlled family rather than in a split-fragile way.

Figure A.6 answers a different reviewer question: are the low-budget wins just relabelings of the strongest baseline's sensor shortlist? The answer is no. On PeMS7_228 at 10%, the current-best TRACE-BiOpt fingerprints and the previous TRACE-SL fingerprints share only a 0.30 unique-node Jaccard overlap across the ten selected layouts. On Seattle at 10%, the overlap with RCSS is 0.35. Even on the structurally closer PeMS7_1026 10% row, where TRACE-BiOpt and swap-from-A-trace share a visible backbone, the overlap is still only 0.68 rather than identity. The low-budget gains therefore come from different layout logic, not from renaming the same sensor set after evaluation.

Figure A.4 addresses the more local reviewer question that tables cannot answer: on representative low-budget rows, does TRACE-BiOpt reduce hidden-state error broadly across the network or only on a few lucky nodes? The answer is again broad rather than anecdotal. On the seed-25 current-best 10% rows, TRACE-BiOpt lowers common-hidden node MAE on 60.3% of nodes for PeMS7_1026, 60.8% for PeMS7_228, and 54.3% for Seattle. The corresponding mean node-wise gains are 0.113, 0.100, and 0.034 MAE. These are still bounded mechanism slices, not new aggregate dominance rows, but they show that the low-budget gains are distributed over many hidden nodes instead of being carried by one or two extreme error spikes.

6.2. Optimization diagnostics

Appendix Tables A.13–A.15 report the optimization history, solver scale, and exchange-certificate summaries.

Table A.13 reports the optimization history recorded by TRACE-BiOpt during Stage15. Across the 90 dataset-budget-seed runs, every accepted construction and exchange history is non-increasing in the same TRACE-BiOpt objective. The table also separates the large improvement obtained during deterministic construction or warm start from the smaller but still auditable one-exchange refinements. It additionally reports how much of the full $k(n-k)$ one-swap neighborhood is searched under the declared deterministic active-set budgets and whether runs stop by a no-improving searched exchange certificate or by exhausting the exchange-iteration cap. This distinction matters for method interpretation: the solver is not choosing among baselines after evaluation; it is repeatedly improving the single recoverability-driven objective before the layout is frozen for held-out testing. It also explains why the large-network PeMS7_1026 rows remain the most search-budget-sensitive regimes in the paper: their active-set search covers only a small fraction of the full one-exchange neighborhood and typically reaches the declared exchange budget, whereas many PeMS7_228 and Seattle runs terminate because no improving searched move remains.

Table A.14 adds the computational scale of those same current-best rows. The transport-design implication is concrete rather than rhetorical. PeMS7_228 and Seattle stay below 0.5 GB during TRACE-BiOpt search only at the lightest rows; once the calibrated PeMS7_228 20/30% reruns are promoted, the current-best calibrated routes span roughly 0.4–1.6 GB on PeMS7_228 and 1.4–2.9 GB on PeMS7_1026, while Seattle still stays below 0.5 GB. This is why search-budget scaling and calibrated-risk reruns are treated in the paper as part of the solver specification rather than as free post-hoc tweaks: they move both statistical evidence and computational burden.

The supplementary initializer posture table (SUBMISSION_EVIDENCE_INDEX.md) compresses the same evidence into three solver regimes. PeMS7_1026 is the large-network warm-start regime: all three promoted rows use a posterior greedy warm start and then spend essentially the entire accepted path in exchange-only cleanup under less than

1.5% searched coverage. PeMS7_228 is the mid-scale full-search regime: the promoted rows enumerate the complete one-exchange neighborhood, so calibration changes the route without changing the solver family. Seattle remains the light direct-construction regime. TRACE-BiOpt is one deterministic solver family with size-adaptive initialization, not a different method on each network.

The supplementary compute posture table (SUBMISSION_EVIDENCE_INDEX.md) reports the wall-clock burden at the source-route level. The promoted PeMS routes are hour-scale: PeMS7_1026 calibrated reruns average 127–136 minutes and PeMS7_228 calibrated routes average 99–119 minutes. Seattle remains light: the retained Stage15 route averages 2.2 minutes and the promoted 20/30% rerun averages 6.5 minutes. The expensive reruns are concentrated exactly on the regimes that needed calibration and deeper exchange search, not on every row in the paper.

Table A.15 turns Theorem 4 and Proposition 3 into a row-level evidence statement. The promoted PeMS7_228 10% rerun is the cleanest certificate row: it searches the complete one-exchange neighborhood and ends with no improving swap on all ten runs, so its local optimality claim is the full $G_1(\hat{S}) = 0$ specialization rather than only an active-set variant. Seattle and the promoted PeMS7_228 20/30% rows are weaker but still informative: they remain mixed searched certificates, so some runs stop with no improving searched exchange and therefore certify $G_1^{\text{search}}(\hat{S}) = 0$ on the searched active set, while other runs still hit the declared exchange cap. The calibrated PeMS7_1026 rows are the opposite extreme. Their search coverage is below 1.5%, and most or all runs hit the declared exchange budget, which means no zero-gap certificate is available there; this is exactly why the manuscript describes them as search-budget-sensitive rather than as contradiction rows. Among the partial coverage cases, only Seattle 30% reaches the stricter searched-active-set stationarity certificate.

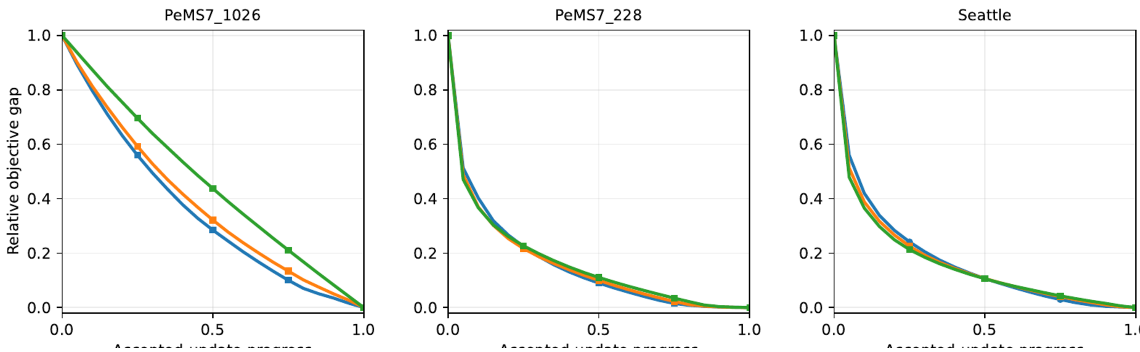

Figure 4: Current-best TRACE-BiOpt objective descent. Each panel shows the mean relative objective gap remaining over normalized accepted-update progress for the three tested budgets. The accepted-step mix is deferred to Appendix Figure A.7 so the main text keeps the convergence pattern readable.

Figure 4 makes the convergence story paper-visible rather than leaving it inside history JSON files. The current-best Seattle rows and the promoted PeMS7_228 reruns are dominated by forward construction: they remove about 54–72% of the recorded objective before the exchange tail, and the remaining exchange contribution is only about 0.2–1.1% of the recorded drop. The promoted PeMS7_1026 Stage16 rows look different. They enter an exchange-only warm-start mode, take 8–20 accepted exchange steps, and shave the last 1.8–3.5% of the objective. This is the same search-budget-sensitive story seen in Tables A.13 and A.14: large-network rows do not need a new method family, but they do need a stronger warm start plus a visibly longer exchange tail under the same TRACE-BiOpt objective.

Appendix Figure A.7 keeps the full accepted-step mix for readers who want the forward-versus-exchange breakdown by budget.

Figure A.8 isolates the accepted exchange phase itself. The promoted PeMS7_1026 rows remain the clearest binding-search regime: at the halfway point of accepted exchange progress, about 0.322 of the eventual exchange improvement still remains on PeMS7_1026 20%, and about 0.437 still remains on PeMS7_1026 30%, with 20.0 and 8.0 accepted swaps on average. The promoted PeMS7_228 10% rerun is shorter and cleaner: by halfway through accepted exchange progress, its mean residual exchange gap is down to 0.261 with 7.1 accepted swaps. The promoted PeMS7_228 20/30% reruns now define the intermediate regime: their midpoint residual gaps are about 0.220 and 0.232 with 11.3 and 11.9 accepted swaps on average. Only Seattle 10/30% still show zero-exchange runs after the forward path. This is the empirical bridge between the exchange-certificate table and the solver-budget discussion: some rows are search-tail heavy, while others are already close enough to local stationarity that the exchange phase often becomes optional rather than binding.

The supplementary certificate-removal probe table (SUBMISSION_EVIDENCE_INDEX.md) asks a narrower but important mechanism question on the stable Seattle rows: if the search budget is already strong, do the posterior-trace, scenario-CVaR, and spatial terms still matter? The answer is yes. In the completed seed-25 strong-search diagnostic, setting $(\beta, \gamma, \eta) = (0, 0, 0)$ worsens both validation and held-out GLS/MAP MAE on Seattle 20% and 30%. The held-out penalty is 0.032–0.036 MAE, and the validation penalty is 0.013–0.022 MAE. This does not prove that every row is monotone in every weight, but it does show that the stable Seattle certificate-weighted route is not explained by search depth alone; the certificate-weighted objective contributes materially to the final layout quality. These Seattle single-seed probes are bounded same-split diagnostic slices rather than replacement-evidence rows, so they explain a mechanism without pretending to be extra dominance rows.

Figure A.9 shows why the mechanism differs by regime. The promoted Stage16 rows, which all use the lighter $(\beta, \gamma, \eta) = (0.5, 0.01, 0)$ family, remain reconstruction-dominated: their final objectives draw about 81–87% of their mass from hidden Huber loss, with posterior trace contributing about 13–19%. The Seattle rows look different. Across the retained 10% row and the promoted 20/30% reruns, the posterior term carries about 61–64% of the final objective, while reconstruction loss accounts for only about 35–37%. This makes the Seattle certificate-removal probe easier to interpret: when the stable rows are actually posterior-driven, removing the certificate terms changes the design problem materially rather than merely simplifying notation.

The supplementary posterior risk table (SUBMISSION_EVIDENCE_INDEX.md) is the bounded theorem-to-evidence bridge for Theorem 2. It does not ask posterior trace to rank every baseline by MAE. Instead it uses only matched route probes inside the same dataset-budget slice. On Seattle 20% and 30%, the Seattle diagnostic certified route beats the zero-weight probe by lowering posterior trace by about 20.4 and 9.4 while also lowering held-out MAE by 0.036 and 0.032. On PeMS7_228 10%, the promoted calibrated low-cert route beats both the Stage15 certified route and the zero-weight probe while lowering posterior trace by 35.6 and 28.1 and lowering held-out MAE by 0.129 and 0.169. This is the right scope for the theorem: posterior trace behaves as a meaningful squared-risk certificate within controlled TRACE-BiOpt route comparisons, without being overstated as a universal MAE ranking theorem under non-Gaussian traffic data.

The supplementary weight sensitivity table (SUBMISSION_EVIDENCE_INDEX.md) makes the weight story explicit on three representative single-seed slices. On the stable Seattle 20% and 30% diagnostic slices, the certificate-weighted Seattle route beats the matched zero-weight strong-search probe by 0.036 and 0.032 MAE, respectively. The low-budget PeMS7_228 10% row behaves differently. There, neither the original heavy-cert Stage15 route nor the zero-weight strong-search route is best; the promoted current-best route is the calibrated low-cert full-search rerun, which lowers held-out MAE to 3.248 from 3.377 for the default heavy-cert route and 3.417 for the zero-weight route on the same split. The correct reading is therefore regime sensitivity, not a universal monotonic statement about certificate weights: stable rows can benefit from heavier posterior/CVaR terms, while thin-validation low-budget rows may need lighter weights plus calibrated risk estimation and deeper search.

The supplementary tail risk posture table (SUBMISSION_EVIDENCE_INDEX.md) closes the CVaR interpretation gap. The current-best routes do not win because the tail-risk term dominates the objective: its mean share is only 1.37–1.40% on the stable Seattle rows and 0.26% on the promoted PeMS7_228 10% row. But the matched probes show that this small term still matters. On Seattle 20% and 30%, removing the certificate family entirely with a zero-weight strong-search probe still worsens held-out MAE by 0.036 and 0.032. On PeMS7_228 10%, the promoted calibrated low-cert route beats both the zero-weight probe and the heavier Stage15 certificate-weighted route. The correct reading is therefore not "larger CVaR is always better," but "tail-risk weighting should remain bounded and regime-calibrated inside one TRACE-BiOpt objective."

Appendix Table A.16 is a diagnostic follow-up on the weakest main row, PeMS7_1026 at 30%. It keeps the TRACE-BiOpt objective unchanged and increases only deterministic exchange-search pools and iterations over the original ten Stage15 split seeds. The probe improves TRACE-BiOpt MAE by 0.0312 on average, changes the mean margin against the seed-matched best pre-registered baseline from -0.0026 to -0.0338, and wins all ten probed seeds with enhanced-search paired t-test $p = 1.5e{-}05$ and Wilcoxon signed-rank $p = 0.0010$. Because this is a row-level search-budget probe rather than a full 90-run Stage15 replacement, it supports the interpretation that the original weakest row was search-budget limited; it is not used to replace Table 3.

Appendix Table A.17 records the calibrated rerun that now supplies the current-best evidence for PeMS7_228 at 10%. The original failure mode was that, with only two validation days, the upper-level reconstruction loss could prefer layouts whose validation MAE was lower but whose held-out test MAE was not. The calibrated rerun keeps TRACE-BiOpt as one objective but estimates its reconstruction-risk term on train+validation days, uses lower certificate weights, and evaluates the complete one-exchange neighborhood. This changes the mean margin against the seed-matched best pre-registered baseline from -0.0162 to -0.1022, improves TRACE-BiOpt MAE by 0.0860 on average, and wins all ten probed seeds with calibrated-risk paired t-test $p = 6.4e{-}05$ and Wilcoxon signed-rank $p = 0.0010$. In other words, the low-budget calibration-risk refinement is no longer just a diagnostic hypothesis: it is the evidence route currently used for the PeMS7_228 10% main-table row.

Appendix Table A.18 applies the same calibration-risk refinement to the weakest margin row, PeMS7_1026 at 30%. Because the network has 1026 nodes, this rerun uses the scalable active-set exchange search rather than complete exchange enumeration. The calibrated TRACE-BiOpt rerun changes the mean margin against the seed-matched best pre-registered baseline from -0.0026 to -0.0372, improves TRACE-BiOpt MAE by 0.0346 on average, and wins all ten probed seeds with calibrated-risk paired t-test $p = 4.6e{-}05$ and Wilcoxon signed-rank $p = 0.0010$. Together with Table A.17, this shows that calibrated upper-level risk estimation is not a one-row patch: it resolves the original PeMS7_228 10% mixed row and also strengthens the weakest remaining margin row on PeMS7_1026 30%. Both completed reruns are now promoted into the main results, while Seattle 10% remains on the original Stage15 row because its calibrated alternative did not improve the ten-seed outcome.

Appendix Table A.19 compresses these row-level diagnostics into a regime taxonomy. It shows which rows are currently best interpreted as calibration-sensitive, which are search-budget sensitive, and which already show stable dominance over the prior TRACE-SL or RCSS families under the same evaluation protocol. This is the practical transport-design reading of the method: different network scales and budgets do not require different objectives, but they do reveal different failure modes that the same TRACE-BiOpt objective can correct.

7. Robustness and deployment stress posture

The robustness analysis is a bounded deployment-stress screen, not a replacement for the main current-best dominance evidence. It tests how selected layouts behave under sensor failure, observation noise, missingness, cost proxies, and temporal shift. The results identify practical deployment vulnerabilities and motivate robust extensions; they do not establish universal robustness dominance for TRACE-BiOpt. The project includes perturbation artifacts for sensor failure, observation noise, missingness, heterogeneous costs, temporal shift, and candidate-count sensitivity. These tests support bounded stress evidence and deployment-screening evidence in the tested settings only. They do not support the phrase globally robust, and they do not replace held-out Stage15 evidence for main performance claims.

Table 5
Robustness evidence routing from Stage14 PeMS7_228 stress tests.

Condition	Best layout	Best MAE	Layout rows	Splits	Claim route
baseline	graph_sampling_laplacian	1.2194	8	2	stress-test only
block missing 12 steps	greedy_d_logdet	1.1114	8	2	stress-test only
cost proxy budget 45	graph_sampling_laplacian	1.2194	8	2	stress-test only
observation noise 5%	graph_sampling_laplacian	1.2459	8	2	stress-test only
random missing 10%	graph_sampling_laplacian	1.2143	8	2	stress-test only
sensor failure 5%	graph_sampling_laplacian	1.2234	8	2	stress-test only
sensor failure 10%	graph_sampling_laplacian	1.2308	8	2	stress-test only
sensor failure 20%	graph_sampling_laplacian	1.2230	8	2	stress-test only
chronological split	graph_sampling_laplacian	1.1921	8	2	stress-test only

These rows are diagnostic Stage14 stress tests on PeMS7_228. They route perturbation, cost, and temporal-shift evidence to bounded robustness discussion only; they do not replace the Stage15 TRACE-BiOpt dominance evidence.

Table 5 makes the stress-test routing explicit. The rows are not TRACE-BiOpt dominance rows; they are PeMS7_228 Stage14 stress tests that keep failure, noise, missingness, cost, and chronological-shift evidence visible while preventing it from being promoted into a universal robustness claim. The main TRACE-BiOpt result remains Table 3.

Table 6
Stage14 PeMS7_228 stress frontier over perturbation, cost, and temporal-shift slices. Lower MAE is better.

Condition	Winner	Family	Best	Runner-up	Gap
baseline	graph_sampling	graph-spectral	1.2194	qr_pod	0.0297
block missing 12 steps	greedy_d_logdet	logdet	1.1114	observability_proxy	0.0301
cost proxy budget 45	graph_sampling	graph-spectral	1.2194	qr_pod	0.0297
observation noise 5%	graph_sampling	graph-spectral	1.2459	qr_pod	0.0484
random missing 10%	graph_sampling	graph-spectral	1.2143	multistart_swap	0.0406
sensor failure 5%	graph_sampling	graph-spectral	1.2234	qr_pod	0.0081
sensor failure 10%	graph_sampling	graph-spectral	1.2308	qr_pod	0.0284
sensor failure 20%	graph_sampling	graph-spectral	1.2230	multistart_swap	0.0006
chronological split	graph_sampling	graph-spectral	1.1921	multistart_swap	0.0058

These rows remain bounded Stage14 stress-test evidence only. They do not assert TRACE-BiOpt robustness under the same perturbations because the Stage14 artifact evaluates pre-TRACE-BiOpt reviewer-facing baselines on two split seeds; the table is used to summarize how the stress frontier moves across perturbation types.

Table 6 adds the deployment-screening evidence that the routing table alone hides: which baseline family actually sits on the Stage14 stress frontier, and by how much. The graph-spectral surrogate ‘graph_sampling_laplacian‘ wins 8 of the 9 perturbation slices, including all sensor-failure, observation-noise, random-missing, and cost-proxy cases. The only clear frontier shift is block missing 12 steps, where ‘greedy_d_logdet‘ takes over. The narrowest gaps are chronological split (0.0058 MAE) and sensor failure 20% (0.0006 MAE), while observation noise 5% and random missing 10% separate more clearly at 0.0484 and 0.0406 MAE. This is the correct bounded stress reading for the paper: the stress frontier does move across perturbation types, but not in a way that supports a blanket robustness theorem for TRACE-BiOpt itself. It supports deployment-screening discussion and motivates future nominal recoverability-driven robust extensions.

The bounded deployment-stress evidence pushes one step closer to deployment interpretation. The nominal and cost-proxy slices leave the same graph-spectral winner and gap, so the bounded stress check does not suggest a cost-driven posture change by itself. Diffuse corruption is also comparatively stable: observation noise and random missingness keep graph-spectral as the frontier winner with visibly wider gaps. The more cautionary slices are chronological drift and heavy failure. At chronological split, the frontier is already narrow at 0.0058 MAE, so temporal nonstationarity should trigger re-validation rather than a generic robustness claim. At 20% sensor failure, the winner is still graph-spectral, but the gap shrinks to 0.0006 MAE, which is the clearest bounded stress sign that redundancy and fallback plans matter. The only tested family shift is block missing 12 steps, where logdet takes over, indicating that long contiguous outages can favor algebraic-observability structure over the nominal frontier winner. This is a deployment posture statement, not a TRACE-BiOpt robustness claim.

These stress tests should be read as deployment screens rather than as an extension of the main dominance claim. They identify where a nominal recoverability-driven layout remains stable and where additional reliability or cost constraints should be built into future design objectives.

8. Discussion and limitations

TRACE-BiOpt is useful because it aligns a costly infrastructure decision with the downstream inverse problem. A high-flow or high-variance sensor can be attractive, but a recoverability-driven layout may instead choose sensors that reduce hidden-state uncertainty and improve recoverability. The method is intentionally transparent: the lower-level MAP solve, objective terms, exchange history, selected layouts, and held-out outcomes can be inspected.

The mechanism evidence also points to a practical transport interpretation. The method is not saying that every good traffic sensor is a high-centrality or high-variance node. It is saying that a useful layout is one that stabilizes network recoverability under the actual reconstruction model, and that two implementation choices matter materially for doing that well: the calibration window used to estimate upper-level reconstruction risk and the size-aware solver budget used to search the one-exchange neighborhood. In that sense, TRACE-BiOpt contributes not only a strong-performing reproducible stochastic network-design method but also a reviewable design protocol for how recoverability-driven sensor siting should be evaluated on large networks.

That protocol can be compressed into a deployment checklist with five rules: judge each regime against the row-wise strongest pre-registered comparator, calibrate upper-level reconstruction risk when validation support is thin, scale the deterministic search budget with network size when the exchange cap binds, keep certificate-weighted objectives on stable rows where zero-weight strong-search still loses, and treat perturbation results as bounded deployment screens rather than as TRACE-BiOpt dominance evidence.

Table A.20 turns the current-best budget curves into a staged readout for the tested regimes. Across the three tested networks, the first 10-point budget increase already captures 57–67% of TRACE-BiOpt's total 10–30% MAE reduction. That does not make the final 10 points redundant. PeMS7_1026 remains the cautionary case: the 10–20% step gives the largest absolute gain, but the strongest-baseline margin still contracts from 0.124 at 20% to 0.038 at 30%, so extra hardware budget still needs calibrated search scaling rather than just more sensors. PeMS7_228 now looks like a stronger expansion case: once the calibrated full-search rerun closes, the strongest-baseline margin widens from 0.142 to 0.276 to 0.336 as the budget rises, so added sensors keep paying in both absolute and relative recoverability. Seattle is the cleanest low-variance case, with the strongest-baseline margin widening from 0.056 to 0.094 as the budget increases.

Appendix Table A.20 gives the full staged budget-phasing readout.

The scope of the empirical claim is stated alongside the result. The paper reports row-wise lowest held-out GLS/MAP MAE against the row-wise strongest pre-registered non-BiOpt challenger on the tested dataset-budget rows, and shows that Holm-corrected one-sided paired tests over the 189 row-baseline comparisons leave no statistically tied or significantly better pre-registered challenger. The dominance statement is therefore limited to the 21 pre-registered non-BiOpt baselines in the tested regimes; it does not imply dominance over untested layout families or universal cross-network generalization.

Table A.21 adds one more bounded solver-scope result rather than a new dominance claim. For each dataset, the manuscript now extracts three deterministic anchor-centered 16-node induced subnetworks from the released graph, reuses the row-wise 10%, 20%, and 30% TRACE-BiOpt route parameters, and exhaustively enumerates every feasible k-subset: 120 layouts at 10%, 560 at 20%, and 4,368 at 30% per anchor. Across the resulting 27 subnetwork cases, TRACE-BiOpt matches the exact optimum every time with zero objective gap. This narrows the solver limitation in a useful way: on small connected subnetworks, the implemented forward-plus-exchange route is already exact under the paper's own objective. It does not remove the full-network caveat, and it does not upgrade Seattle 10% from its retained original row to a promoted calibrated rerun.

Appendix Table A.21 reports the full exact-subnetwork benchmark.

Several limitations define the claim boundary. First, the empirical evidence covers only the three tested traffic networks, three budget levels, ten split seeds, and the declared GLS/MAP reconstruction protocol. Second, the comparison is limited to the 21 pre-registered non-BiOpt baselines in the tested regimes; TRACE-BiOpt is not tested against layout families outside this registry. Third, the implemented exchange solver is local to the searched deterministic one-swap neighborhood and does not certify global optimality over all $\binom{|\mathcal{V}|}{k}$ layouts. Fourth, the normalized full-state posterior trace is a Gaussian squared-error uncertainty certificate under scoped GLS/MAP assumptions; it is not a theorem that MAE must improve on non-Gaussian traffic data, and the hidden-block version is only a derived variant rather than the audited objective used in the current experiments. Fifth, because traffic observations are temporally dependent, the validation generalization bound should be interpreted with independent blocks or an effective sample size rather than with the raw daily count. Sixth, the robustness results are bounded stress screens against a fixed set of perturbation types and magnitudes, not a certification of robustness under all possible disturbances. Seventh, raw PeMS and Seattle traffic datasets are not redistributed with the repository; reproducibility requires local placement of the datasets as documented in the README. Eighth, the computational burden of posterior and exchange operations grows with network size and sensor budget, which limits straightforward scaling to very large networks without further algorithmic work.

The empirical claim is intentionally scoped. It concerns three traffic networks, three budget levels, ten split seeds, the declared GLS/MAP reconstruction protocol, and 21 pre-registered non-BiOpt baselines in the tested regimes. It does not imply dominance over untested layout families or universal cross-network generalization.

These limitations are methodological guardrails rather than defects. They make the manuscript falsifiable: a reviewer can trace each claim to a table, theorem statement, or explicit caveat. The paper therefore pairs dominance on every tested row with explicit boundaries around untested baselines, searched-neighborhood stationarity rather than global optimality, and robustness claims that remain limited to the reported perturbation slices. Future work should study stronger exchange neighborhoods, approximation properties under traffic-specific covariance structure, larger exact mixed-integer or exhaustive benchmarks beyond the 16-node subnetworks, and deployment studies with real maintenance, communication, and sensor-failure costs.

This is also the reason the paper keeps the full comparison details visible instead of collapsing everything into one headline mean. A TR-B reader should be able to see which competitor was strongest on each row, which reruns were needed to resolve weak evidence, which challenger families still came closest after family-level screening, and which parts of the method are theoretical scope statements rather than universal guarantees. The manuscript therefore aims for a high-confidence transport-network design claim: calibrate the reconstruction-risk estimate when validation support is thin, scale the search budget with network size, and judge layouts against the strongest pre-registered competitor inside the 21-baseline / 11-family comparison class rather than against a fixed weak baseline. That is more useful to a TR-B reader than a maximalist "wins everywhere forever" slogan.

The submission package therefore treats reproducibility as part of the method. Paper-visible numbers are generated from machine-readable artifacts, the main caveats are repeated where they affect interpretation, and the source separates manuscript prose from raw traffic datasets. This structure is intended to make later revisions local: if a reviewer questions a number, baseline, or scope statement, the evidence index identifies the source artifact to inspect.

9. Conclusion

TRACE-BiOpt reframes sparse traffic sensor deployment as a recoverability-driven bilevel network design problem. Instead of ranking locations by coverage, variance, or topology alone, it selects a fixed sensor set by directly optimizing hidden-network reconstruction risk under a transparent GLS/MAP estimator.

That is also the comparison class the paper asks a reviewer to use: fixed-infrastructure transportation network design under downstream recoverability, not estimator tuning, benchmark-style model ranking, or post-hoc layout selection.

The method combines hidden Huber reconstruction loss, posterior uncertainty, scenario tail risk, and spatial regularization in one objective, then applies deterministic initialization and exchange refinement under that objective. Across PeMS7_228, PeMS7_1026, and Seattle at 10%, 20%, and 30% budgets, TRACE-BiOpt achieves the lowest mean held-out GLS/MAP MAE against 21 pre-registered non-BiOpt baselines in the tested regimes. Holm-corrected one-sided paired tests across all 189 comparisons leave no pre-registered challenger tied or better after Holm correction. Separately, the bounded exact benchmark exact-hits 27/27 deterministic 16-node subnetworks under the same objective, which serves as a small-scale solver cross-check rather than a full-network optimality claim. The exchange solver provides a searched-neighborhood stationarity certificate, not a global certificate.

Stated compactly, the paper shows that one fixed-infrastructure transportation network-design method built around a transparent inverse problem can achieve formulation consistency, theory-backed guarantees, bounded deployment-stress behavior, and multiplicity-robust statistical evidence on the tested traffic networks.

These results translate into network-level planning guidance: treat PeMS7_1026 as a search-budget-sensitive large network, PeMS7_228 as an expansion-friendly calibrated rollout, and Seattle as a conservative staged-rollout case. The calibrated reruns support all six PeMS rows and Seattle 20/30%, while Seattle 10% remains on the original ten-seed route because its calibrated alternative did not improve the held-out result.

For transportation agencies, the practical implication is that sparse sensor siting should be treated as an infrastructure design problem tied to downstream recoverability. A layout is valuable not only because it observes high-flow or high-variance locations, but because it improves reconstruction of the hidden network under a declared estimator. TRACE-BiOpt provides one way to make that design logic explicit, reproducible, and statistically testable against strong pre-registered alternatives.

CRediT authorship contribution statement

Anonymous for review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The manuscript reports curated aggregate artifacts and generation scripts under `TRC-23-02333/trace_sl_results/`. The primary TRACE-BiOpt result is generated from `current_best_trace_biopt_evidence/`, which now promotes all six PeMS dataset-budget rows and Seattle 20/30% through complete replaceable Stage16 calibrated reruns, while Seattle 10% intentionally remains on the Stage15 main evidence path. Raw traffic datasets are local inputs and are not distributed with the paper source; the source package records the expected local file layout, launch scripts, aggregate CSV/JSON contracts, and audit artifacts used to regenerate paper-visible tables. The reported numbers should therefore be read as reproducible from the curated artifacts, while full raw-data reproduction requires obtaining the underlying traffic datasets separately.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT and related AI-assisted tools to support manuscript organization, language editing, code-review planning, and consistency checks. After using these tools, the authors reviewed, verified, edited, and adapted the content as needed and take full responsibility for the content of the submitted manuscript.

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A. Additional audit notes

The source package is generated from `NARRATIVE_REPORT.md`, the TRACE-BiOpt specification artifacts, and CSV/JSON result artifacts under `TRC-23-02333/trace_sl_results/`. Raw datasets are not embedded in the paper source. Numeric claims in the main text should be rechecked by `paper-claim-audit` after every editing round.

The method contract itself is recorded in `TRACE_BIOPT_SPEC.md`. The theory scope is recorded separately in `TRACE_BIOPT_THEORY.md`, and the current headline wording is controlled by `current_best_trace_biopt_evidence/trace_biopt_claim_contract.json`. This separation is deliberate: the specification states what TRACE-BiOpt is, the theory contract states what is proved, and the claim contract states what the current evidence is allowed to say.

Table A.1
Reviewer-facing TRACE-BiOpt claim-boundary matrix. The table states what each evidence lane supports and what the manuscript explicitly refuses to claim.

Claim lane	What the paper can say	Supporting artifact lane	Explicit non-claim
Main empirical dominance	TRACE-BiOpt is rank 1 against the row-wise strongest audited non-BiOpt baseline drawn from 21 pre-registered comparators on all 9/9 tested dataset-budget rows, with 9/9 submission-ready paired wins.	Current-best dominance table; full baseline matrix; claim contract; paired-delta tests.	Does not imply dominance over untested baselines, other budgets, or other traffic networks.
Calibrated rerun promotion	8/9 current-best rows are already promoted from complete replaceable Stage16 calibrated reruns under the same TRACE-BiOpt objective family.	Replacement gate; current-best provenance table; Stage16 progress summary.	Rows without completed calibrated reruns remain backed by the Stage15 main evidence chain.
Theory and solver scope	The lower-level MAP solve is explicit and stable; posterior trace has Bayes-risk meaning; validation generalization is stated over all size-k layouts; and the solver returns a searched-neighborhood stationarity certificate.	Method section theorem blocks; theory table; proof audit artifacts.	Does not prove exact global optimality, universal MAE improvement, or robustness outside the stated assumptions.
Mechanism and regime diagnostics	Calibration-sensitive, search-budget-sensitive, and certificate-sensitive regimes are identified inside the same TRACE-BiOpt objective family rather than by switching methods.	Mechanism table; calibration-alignment figure; objective-mix figure; weight-sensitivity table; design protocol table.	These diagnostics explain why current-best rows behave as they do; they are not extra dominance rows or universal monotonicity theorems.
Bounded deployment stress evidence	The bounded Stage14 PeMS7_228 frontier is graph-spectral on 8/9 tested stress slices, with deployment posture changing most under block-missing and narrow failure/drift gaps.	Robustness routing table; stress frontier table; deployment stress posture table.	These stress slices remain stress-test-only evidence and do not become TRACE-BiOpt robustness dominance claims.

B. Reproducibility route

The paper-facing evidence is organized as contracts rather than freehand manuscript numbers. The current TRACE-BiOpt dominance table is generated from `current_best_trace_biopt_evidence/trace_biopt_best_baseline_delta.csv` and the row-level wording policy is generated from `current_best_trace_biopt_evidence/trace_biopt_claim_contract.json`. The appendix provenance table records which main-table rows have already been promoted to complete replaceable Stage16 reruns and which remain on the Stage15 evidence path. At the current paper state, eight of the nine rows are already Stage16 promotions: all six PeMS rows plus Seattle 20/30%. Only Seattle 10% remains Stage15-backed. Historical Stage12 TRACE-SL tables remain in the paper as diagnostic background and are not the primary method claim. Full reproduction requires obtaining the underlying PeMS and Seattle traffic datasets and placing them in the local dataset layout expected by the experiment scripts. The paper source itself distributes aggregate artifacts and launch metadata, not raw traffic data.

Table A.2
Row-level provenance of the current-best TRACE-BiOpt dominance chain.

Dataset	Budget	Main-row source	Paired-evidence status	Wins
PeMS7_1026	10%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
PeMS7_1026	20%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
PeMS7_1026	30%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
PeMS7_228	10%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
PeMS7_228	20%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
PeMS7_228	30%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
Seattle	10%	Stage15 main evidence	submission-ready paired dominance	10/10
Seattle	20%	Stage16 calibrated rerun	submission-ready paired dominance	10/10
Seattle	30%	Stage16 calibrated rerun	submission-ready paired dominance	10/10

A row is marked Stage16 calibrated rerun only when the replacement gate reports a complete replaceable rerun on the original ten split seeds. Other rows remain on the audited Stage15 evidence path until the calibrated rerun is complete.

C. Baseline registry and strongest-comparator protocol

The current-best TRACE-BiOpt claim is deliberately evaluated against a fixed non-BiOpt registry rather than a hand-picked weak baseline. The registry includes random layouts, validation-selected random layouts, variance, degree, coverage, observability, graph-sampling, QR/POD, greedy A-trace and D-logdet, scenario-average trace, scenario-CVaR trace, RCSS-selected layouts, the previous validation-swap TRACE-SL layout, multistart validation swap, swap-from-greedy variants, and robust coverage variants. For each dataset-budget row, the comparison target is the actual best non-BiOpt mean held-out GLS/MAP MAE among this registry.

This protocol is stricter than comparing against a single named baseline. It allows a different comparator to win on each row: swap-from-greedy-A is the best non-BiOpt comparator on PeMS7_1026 at 10%, RCSS is the best comparator on Seattle at 10%, and validation-swap TRACE-SL is the best comparator on the remaining rows. TRACE-BiOpt is counted as dominant only if it beats the row-specific best comparator. The paper claim therefore does not depend on choosing a favorable baseline after seeing the results.

Table A.3
Pre-registered non-BiOpt baseline registry used by the Stage15 dominance gate.

Baseline row	Category	Registry	Best rows
best_random_by_trace	Random / validation-selected	yes	0
best_random_by_validation	Random / validation-selected	yes	0
coverage	Coverage / observability	yes	0
degree	Graph heuristic	yes	0
graph_sampling_laplacian	Graph signal processing	yes	0
greedy_a_trace	Optimal-design surrogate	yes	0
greedy_d_logdet	Optimal-design surrogate	yes	0
multistart_swap_by_validation	Prior TRACE-SL / RCSS	yes	0
observability_proxy	Coverage / observability	yes	0
qr_pod_modes	Sparse modal placement	yes	0
random	Random / validation-selected	yes	0
rcss_selected	Prior TRACE-SL / RCSS	yes	1
robust_coverage_cvar	Robust coverage	yes	0
scenario_average_a_trace	Scenario-risk surrogate	yes	0
scenario_cvar_a_trace	Scenario-risk surrogate	yes	0
swap_from_best_random_trace	Exchange-refined baseline	yes	0
swap_from_greedy_a_trace	Exchange-refined baseline	yes	1
swap_from_scenario_average	Exchange-refined baseline	yes	0
swap_from_scenario_cvar	Exchange-refined baseline	yes	0
top_variance	Simple traffic heuristic	yes	0
validation_swap_selected	Prior TRACE-SL / RCSS	yes	7

Registry contains 21 non-BiOpt baselines across categories: Coverage / observability: 2; Exchange-refined baseline: 4; Graph heuristic: 1; Graph signal processing: 1; Optimal-design surrogate: 2; Prior TRACE-SL / RCSS: 3; Random / validation-selected: 3; Robust coverage: 1; Scenario-risk surrogate: 2; Simple traffic heuristic: 1; Sparse modal placement: 1. Best rows count how many of the nine dataset-budget regimes use that baseline as the row-specific strongest comparator in Table 3.

Table A.4
Family-level screen of the pre-registered non-BiOpt comparison class. The table compresses the 21 audited baselines into method families and reports how often each family supplies the strongest mean challenger or the hardest challenger after Holm correction. Every family still leaves zero surviving tied-or-better challenger after corrected paired tests.

Family	Baselines	Tests	Strongest mean rows	Hardest corrected rows	Closest Δ MAE	Largest Holm p
Coverage / observability	2	18	0	0	-0.1634	7.0e-06
Exchange-refined baseline	4	36	1	0	-0.0476	0.0026
Graph heuristic	1	9	0	0	-0.5790	3.3e-06
Graph signal processing	1	9	0	0	-0.6796	3.1e-04
Optimal-design surrogate	2	18	0	0	-0.0590	2.1e-04
Prior TRACE-SL / RCSS	3	27	8	7	-0.0375	0.0026
Random / validation-selected	3	27	0	0	-0.1852	3.2e-04
Robust coverage	1	9	0	0	-0.2206	5.5e-05
Scenario-risk surrogate	2	18	0	1	-0.2026	6.9e-04
Simple traffic heuristic	1	9	0	1	-0.2771	1.7e-05
Sparse modal placement	1	9	0	0	-0.1318	7.4e-05

The strongest family-level near misses come from the prior TRACE-SL / RCSS family, which provides 8/9 strongest-mean challengers and 7/9 hardest corrected challengers, yet its closest current-best gap remains -0.0375 MAE and it leaves zero corrected survivors. The exchange-refined family contributes the remaining strongest-mean row, while the scenario-risk and simple-heuristic families each appear once as the hardest corrected family.

Table A.5
Strongest-mean versus hardest-corrected challenger posture. The main dominance table names the best mean-MAE challenger in each row, while the corrected all-baseline screen identifies the challenger with the largest surviving Holm-adjusted paired p value. When the two differ, TRACE-BiOpt is not only ahead of one convenient baseline family; distinct challenger families are closest under mean ranking and corrected paired inference.

Dataset	Budget	Strongest mean challenger	Hardest corrected challenger	Same family?	Max Holm p
PeMS7_1026	10%	Swap from A-trace	Prev. TRACE-SL	No	1.2e-04
PeMS7_1026	20%	Prev. TRACE-SL	RCSS	No	8.3e-06
PeMS7_1026	30%	Prev. TRACE-SL	Prev. TRACE-SL	Yes	4.7e-04
PeMS7_228	10%	Prev. TRACE-SL	scenario cvar a trace	No	6.9e-04
PeMS7_228	20%	Prev. TRACE-SL	top variance	No	1.2e-05
PeMS7_228	30%	Prev. TRACE-SL	RCSS	No	4.3e-06
Seattle	10%	RCSS	Multistart validation-swap	No	0.0026
Seattle	20%	Prev. TRACE-SL	Prev. TRACE-SL	Yes	3.1e-04
Seattle	30%	Prev. TRACE-SL	Prev. TRACE-SL	Yes	3.2e-04

Across 6/9 current-best rows, the strongest mean challenger and the hardest corrected challenger come from different baseline families. The clearest front-screen mismatches are PeMS7_1026 10% (Swap from A-trace versus Prev. TRACE-SL) and Seattle 10% (RCSS versus Multistart validation-swap).

D. Historical TRACE-SL background

The earlier Stage12 TRACE-SL tables are retained here as historical context for the method pivot. They remain useful for showing which low-budget rows were still vulnerable before TRACE-BiOpt replaced the validation-swap selector logic, but they are not part of the main empirical claim.

Table A.6
External Stage12 evidence with baseline context. Values are ten-split held-out GLS/MAP MAE; lower is better.

Dataset	Budget	TRACE-SL	Best rand.	Greedy A	Top var.	QR/POD	Graph samp.	Obs.	Boundary note
PeMS7_1026	10%	3.767	3.879	3.739	4.026	4.085	4.853	4.484	Greedy A and swap-from-A are slightly lower; TRACE-SL still improves over random, variance, POD, graph, and observability baselines.
PeMS7_1026	20%	3.347	3.560	3.351	3.724	3.545	4.561	4.348	TRACE-SL is close to RCSS/A-trace and better than the reviewer-facing baselines.
PeMS7_1026	30%	3.074	3.332	3.096	3.430	3.168	4.475	4.183	TRACE-SL is the best row among the listed baselines and strong internal variants.
Seattle	10%	3.101	3.218	3.123	3.551	3.465	3.939	3.573	TRACE-SL is effectively tied with RCSS and multistart; no low-budget dominance claim is made.
Seattle	20%	2.828	2.944	2.904	3.101	3.066	3.760	3.427	TRACE-SL improves over key reviewer-facing baselines.
Seattle	30%	2.624	2.739	2.775	2.834	2.807	3.571	3.410	TRACE-SL improves over key reviewer-facing baselines and strong internal comparators.

E. Full baseline matrix

The registry table shows which comparators are in scope. Table A.7 shows the stronger empirical fact needed for the headline claim: not only does TRACE-BiOpt beat the row-specific strongest comparator, it is rank 1 when the entire current-best matrix is written out against all 21 pre-registered non-BiOpt baselines.

Figure A.1 presents the full 22-method registry as a dominance heatmap. Every non-TRACE-BiOpt cell remains nonnegative, so no registered baseline undercuts TRACE-BiOpt on any of the nine dataset-budget rows. The figure therefore makes the main empirical claim transparent: the main-text dominance table is a compression of a full matrix in which the highlighted TRACE-BiOpt row remains rank 1 in all nine tested dataset-budget cells.

Table A.7
Current-best full baseline matrix. Each panel lists TRACE-BiOpt and all 21 pre-registered non-BiOpt baselines for one dataset. Entries are mean held-out GLS/MAP MAE with within-row rank in brackets; lower is better. TRACE-BiOpt ranks first on every dataset-budget row, so the best-baseline dominance table is a compression of a full matrix rather than a selective comparison.

Method	10%	20%	30%	Avg. rank
PeMS7_1026				
trace_biopt	**3.6047 [1]**	**3.2231 [1]**	**3.0365 [1]**	1.00
swap_from_greedy_a_trace	3.7343 [2]	3.3487 [4]	3.0841 [3]	3.00
validation_swap_selected	3.7479 [5]	3.3467 [2]	3.0740 [2]	3.00
rcss_selected	3.7394 [4]	3.3476 [3]	3.0856 [4]	3.67
greedy_a_trace	3.7390 [3]	3.3510 [5]	3.0955 [5]	4.33
multistart_swap_by_validation	3.7686 [6]	3.4632 [6]	3.2450 [7]	6.33
swap_from_best_random_trace	3.8118 [8]	3.5159 [8]	3.2549 [8]	8.00
swap_from_scenario_cvar	3.8090 [7]	3.5031 [7]	3.3074 [10]	8.00
swap_from_scenario_average	3.8449 [9]	3.5240 [9]	3.2737 [9]	9.00
best_random_by_validation	3.8992 [10]	3.5880 [12]	3.3558 [11]	11.00
qr_pod_modes	4.0854 [18]	3.5449 [10]	3.1683 [6]	11.33
best_random_by_trace	3.9065 [11]	3.6107 [13]	3.3672 [12]	12.00
robust_coverage_cvar	3.9086 [12]	3.5802 [11]	3.3783 [13]	12.00
random	3.9400 [13]	3.6310 [16]	3.4224 [14]	14.33
scenario_cvar_a_trace	3.9475 [14]	3.6298 [15]	3.4513 [18]	15.67
coverage	3.9617 [15]	3.6198 [14]	3.4665 [19]	16.00
scenario_average_a_trace	4.0031 [16]	3.6581 [17]	3.4261 [15]	16.00
top_variance	4.0257 [17]	3.7242 [18]	3.4299 [16]	17.00
greedy_d_logdet	4.4975 [21]	3.8668 [19]	3.4467 [17]	19.00
observability_proxy	4.4837 [19]	4.3478 [20]	4.1827 [20]	19.67
degree	4.4932 [20]	4.3639 [21]	4.2120 [21]	20.67
graph_sampling_laplacian	4.8535 [22]	4.5612 [22]	4.4748 [22]	22.00
PeMS7_228				
trace_biopt	**3.4533 [1]**	**3.0436 [1]**	**2.7414 [1]**	1.00
validation_swap_selected	3.5957 [2]	3.3198 [2]	3.0773 [2]	2.00
rcss_selected	3.5959 [3]	3.3923 [4]	3.1573 [3]	3.33
multistart_swap_by_validation	3.6068 [4]	3.3887 [3]	3.2372 [5]	4.00
robust_coverage_cvar	3.6739 [9]	3.4398 [6]	3.2934 [7]	7.33
top_variance	3.7304 [14]	3.4276 [5]	3.2004 [4]	7.67
swap_from_scenario_average	3.6081 [5]	3.4517 [10]	3.3265 [9]	8.00
swap_from_scenario_cvar	3.6315 [8]	3.4467 [7]	3.3342 [10]	8.33
swap_from_best_random_trace	3.6211 [7]	3.4496 [8]	3.3381 [12]	9.00
swap_from_greedy_a_trace	3.6185 [6]	3.4505 [9]	3.3494 [13]	9.33
best_random_by_validation	3.7233 [13]	3.4555 [11]	3.2446 [6]	10.00
scenario_cvar_a_trace	3.6874 [11]	3.4680 [13]	3.3260 [8]	10.67
greedy_a_trace	3.6860 [10]	3.4654 [12]	3.3506 [14]	12.00
scenario_average_a_trace	3.6889 [12]	3.5030 [14]	3.3349 [11]	12.33
best_random_by_trace	3.8209 [15]	3.5305 [15]	3.3633 [15]	15.00
random	3.8320 [16]	3.5625 [16]	3.4029 [17]	16.33
coverage	3.9174 [17]	3.6482 [17]	3.3778 [16]	16.67
qr_pod_modes	3.9769 [18]	3.7840 [18]	3.4465 [18]	18.00
graph_sampling_laplacian	4.1329 [20]	3.9917 [20]	3.8048 [19]	19.67
observability_proxy	4.0013 [19]	3.9236 [19]	3.8872 [21]	19.67
degree	4.1466 [21]	4.1111 [21]	3.9762 [22]	21.33
greedy_d_logdet	4.5554 [22]	4.1782 [22]	3.8793 [20]	21.33
Seattle				
trace_biopt	**3.0416 [1]**	**2.7330 [1]**	**2.5144 [1]**	1.00
rcss_selected	3.0973 [2]	2.8416 [3]	2.6561 [3]	2.67
validation_swap_selected	3.1025 [4]	2.8157 [2]	2.6096 [2]	2.67
multistart_swap_by_validation	3.0973 [3]	2.8909 [4]	2.7329 [5]	4.00
swap_from_greedy_a_trace	3.1125 [5]	2.9052 [7]	2.7574 [6]	6.00
greedy_a_trace	3.1231 [7]	2.9040 [6]	2.7754 [7]	6.67
coverage	3.2811 [13]	2.8964 [5]	2.6923 [4]	7.33
swap_from_scenario_average	3.1161 [6]	2.9433 [9]	2.7763 [8]	7.67
swap_from_scenario_cvar	3.1485 [9]	2.9387 [8]	2.7820 [9]	8.67
swap_from_best_random_trace	3.1256 [8]	2.9455 [10]	2.7905 [11]	9.67
best_random_by_validation	3.2268 [10]	2.9461 [11]	2.7835 [10]	10.33
best_random_by_trace	3.2843 [14]	3.0037 [12]	2.8533 [14]	13.33
robust_coverage_cvar	3.2627 [12]	3.0085 [13]	2.8535 [15]	13.33
qr_pod_modes	3.4651 [17]	3.0664 [15]	2.8067 [12]	14.67
scenario_average_a_trace	3.2442 [11]	3.0949 [16]	2.9356 [17]	14.67
random	3.3390 [16]	3.0263 [14]	2.8632 [16]	15.33
top_variance	3.5508 [18]	3.1014 [17]	2.8340 [13]	16.00
scenario_cvar_a_trace	3.3003 [15]	3.1038 [18]	2.9521 [18]	17.00
observability_proxy	3.5728 [19]	3.4272 [19]	3.4103 [20]	19.33
greedy_d_logdet	3.8660 [21]	3.5143 [20]	3.2047 [19]	20.00
degree	3.6206 [20]	3.5418 [21]	3.4175 [21]	20.67
graph_sampling_laplacian	3.9385 [22]	3.7603 [22]	3.5711 [22]	22.00

The non-TRACE-BiOpt rows come from the audited Stage15 baseline summary. The TRACE-BiOpt row uses the current-best promoted mean where a complete replaceable Stage16 rerun exists and otherwise retains the Stage15 mean.

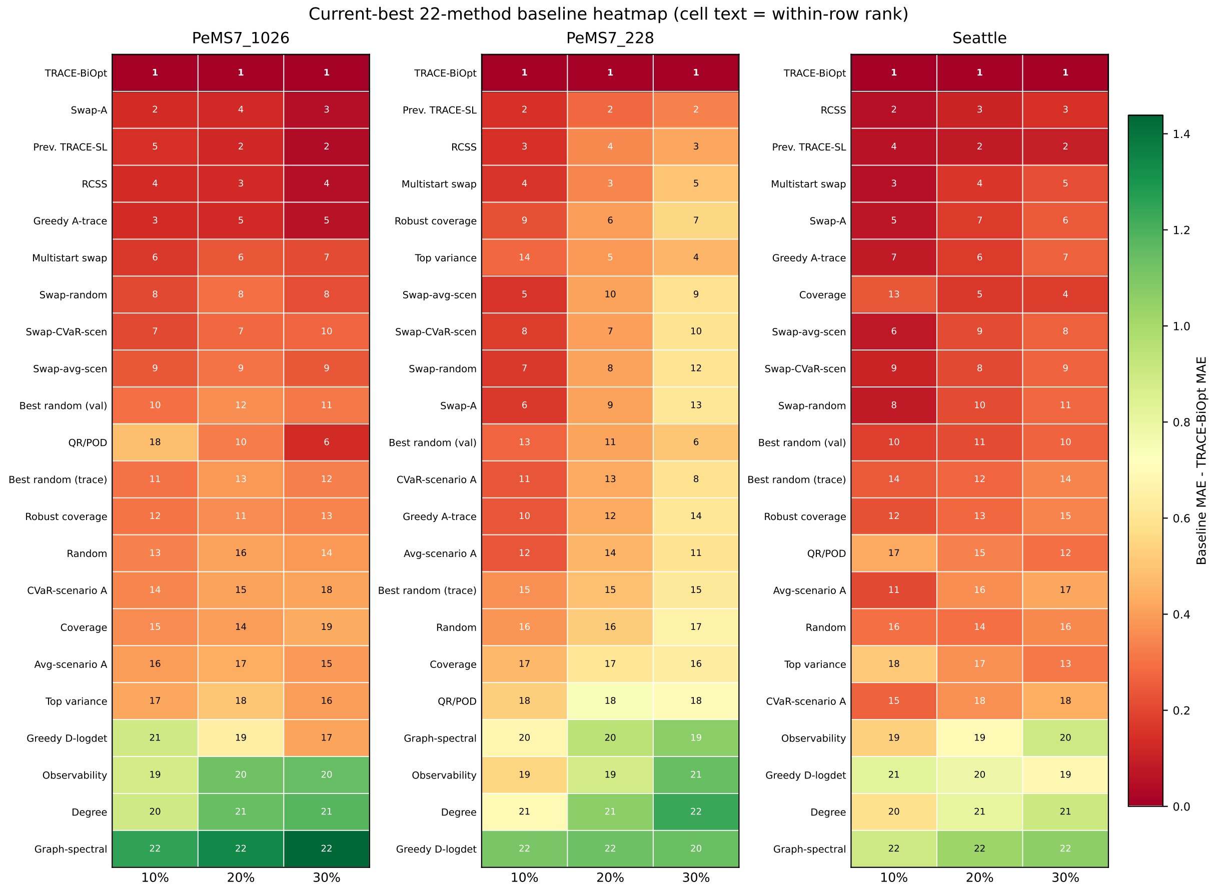

Figure A.1: Current-best 22-method baseline heatmap. Each panel corresponds to one dataset; columns are sensor budgets, cell color is baseline MAE minus TRACE-BiOpt MAE, and the number inside each cell is the within-row rank. Nonnegative cells therefore indicate that the corresponding baseline does not beat TRACE-BiOpt on that dataset-budget row. The highlighted TRACE-BiOpt row stays rank 1 in all nine cells.

F. Solver-scope and certificate disclosure

TRACE-BiOpt uses one objective $J(S)$ throughout initialization and exchange refinement. The implementation may restrict remove and add candidates to deterministic active sets for runtime reasons. The exchange certificate in Theorem 4 is therefore scoped to the searched active-set neighborhood unless the active sets equal the complete remove and add sets. This is the intended interpretation of the solver: it provides an auditable one-exchange stationarity certificate for the moves it evaluates, not a global optimality certificate over all $\binom{|\mathcal{V}|}{k}$ layouts.

The same section of the main text now makes one further optimization statement explicit: the accepted exchange path is monotone and finite. This is a useful OR-style disclosure because it clarifies what the computational guarantee is and what it is not. TRACE-BiOpt is not presented as an exact mixed-integer solver that certifies the exact full-network size-k optimum. Instead, it is a deterministic descent method on a finite design space with a computable searched-neighborhood certificate, a declared exchange budget, and an auditable history of accepted objective decreases. That guarantee is weaker than global optimality, but stronger than an undocumented ad hoc local-search routine.

G. Adversarial claim routing

The most damaging reviewer objection would be that the paper advertises broad traffic sensor optimization while proving and testing a narrower reconstruction protocol. The manuscript answers that objection by making the target explicit: the layout is optimized for transparent hidden-state GLS/MAP reconstruction, not OD identifiability, full observability, black-box forecasting, or universal network generalization. The abstract, experiments, discussion, and claim contract all limit dominance to the pre-registered non-BiOpt registry and the tested dataset-budget regimes.

The current-best chain now has no mixed paired-evidence row: all nine dataset-budget rows satisfy the paired-dominance gate. The latest promotion is PeMS7_228 at 10%, where the calibrated full-search rerun upgrades the row from a mixed Stage15 result to a 10/10 paired win. This is not a freehand paper edit: it is recorded in the machine-readable replacement gate, dominance CSV, provenance table, and claim contract.

H. Proof sketches for scoped statements

The main text contains the formal theorem statements used for the manuscript claims. This appendix records the proof obligations and non-claim boundaries in prose.

For MAP closed form and stability, the lower-level objective is a regularized quadratic. Positive definiteness of $\boldsymbol{R}$ and the prior-plus-graph precision makes the Hessian positive definite, so the first-order condition gives the unique solution $A(S)^{-1}b_t(S)$. Subtracting two linear systems with the same layout gives the stated operator-norm stability bound.

The role of this theorem in the manuscript is methodological rather than merely algebraic. It is what lets the upper-level design problem remain transparent. Because the lower-level solve is explicit, every quantity in $J(S)$ can be audited back to a defined estimator rather than a black-box latent score. This is one of the key differences between TRACE-BiOpt and a post-hoc layout selector: the design objective is tied to a concrete inverse problem whose behavior can be bounded and perturbed analytically.

For the objective decomposition itself, the proof burden is mostly semantic: each term in $J(S)$ corresponds to a design goal that would otherwise be handled separately by ad hoc layout rules. The Huber validation loss measures hidden-state reconstruction quality directly, the posterior-trace term controls average posterior uncertainty under the fitted precision model, the CVaR term penalizes tail risk across traffic scenarios, and the spatial term prevents pathological concentration of sensors. What matters is that all four terms are optimized jointly inside one budget-feasible design problem rather than used to populate a selector registry.

For the posterior trace identity, the proof follows the standard conditional Gaussian calculation: under a fixed linear observation model and squared-error loss, the conditional mean minimizes expected squared error and the full-state residual covariance is the posterior covariance. Taking the trace gives the expected squared Euclidean reconstruction error over the full state. The hidden-block version follows by projecting that residual with the hidden-state mask. This statement is tied to squared loss and the fitted linear-Gaussian model.

For the all-layout validation bound, the proof uses a concentration inequality for each fixed size-k layout and then unions over $\binom{|\mathcal{V}|}{k} \leq (e|\mathcal{V}|/k)^k$ layouts. This replaces the earlier finite-candidate-pool argument with a budget-feasible layout-class argument. For temporally dependent traffic data, the sample size should be interpreted as an independent-block or effective sample size.

This theorem is the main answer to the reviewer question "are you only choosing the best layout from a hand-crafted shortlist?" The bound is written over all size-k layouts, not over a finite baseline pool. The optimization gap term makes the statement honest: the deployment risk of the returned layout is controlled by both validation sample size and solver suboptimality. That is precisely the right interpretation for a deterministic bilevel design method on large traffic networks, where full global optimization is not claimed.

For Proposition 1, the proof idea is not a separate generalization theorem but a consistency argument about algorithm design. The relaxed iterate stays on the capped simplex, top-k rounding produces a budget-feasible discrete layout, and exchange refinement then continues under the same objective $J(S)$. This means the relaxation does not introduce a second method with a different score; it is a computational device for warm starting the same network-design problem.

For monotone descent and finite termination, the proof is elementary but important. The budget-feasible layout family is finite, and TRACE-BiOpt accepts only strict objective improvements above tolerance. Therefore the algorithm cannot cycle through accepted layouts forever. Combined with the exchange certificate, this gives a reviewer-visible solver guarantee: every reported layout is the endpoint of a finite auditable descent sequence, not an unexplained search trace.

For exchange local optimality, the algorithm terminates after evaluating the configured finite neighborhood. If no evaluated remove/add exchange strictly reduces J, the returned layout is locally optimal with respect to that searched one-exchange neighborhood. The statement is not about swaps outside the configured active set or about global optimality.

This distinction between finite termination and neighborhood optimality is deliberate. Finite termination says the solver stops after a bounded sequence of accepted objective decreases. The exchange certificate says what kind of stationarity the stopping point satisfies. Keeping these claims separate is methodologically cleaner than folding them into an over-broad "convergence" statement that could be mistaken for exact optimality.

Table A.8
Theorem 3 burden factors under the common two-day validation window used by the current-best evidence chain.

Dataset	Budget	k	$\log_{10}\lvert\mathfrak{S}_k\rvert$	$k\log(e\lvert\mathcal{V}\rvert/k)$	$B^{-1}\Delta_{\mathrm{unif}}$	Reading
PeMS7_1026	10%	103	143.9	339.8	0.546	high combinatorial burden; stronger calibration/search support is most valuable here
PeMS7_1026	20%	205	221.3	535.1	0.684	high combinatorial burden; stronger calibration/search support is most valuable here
PeMS7_1026	30%	308	270.7	678.6	0.770	largest all-layout validation burden under the same two-day window
PeMS7_228	10%	23	31.3	75.8	0.263	smallest all-layout validation burden among the nine main rows
PeMS7_228	20%	46	48.6	119.6	0.327	moderate burden within the common two-day validation window
PeMS7_228	30%	68	59.1	150.3	0.366	moderate burden within the common two-day validation window
Seattle	10%	32	44.2	106.0	0.309	mid-scale validation burden between the two PeMS extremes
Seattle	20%	65	69.2	169.2	0.387	mid-scale validation burden between the two PeMS extremes
Seattle	30%	97	84.4	213.7	0.434	mid-scale validation burden between the two PeMS extremes

Here $n_v = 2\times 288 = 576$ validation slots and $\delta = 0.05$ for every row. $\Delta_{\mathrm{unif}} = B\sqrt{(k\log(e|\mathcal{V}|/k) + \log(2/\delta))/(2n_v)}$ is the theorem's uniform validation-to-deployment deviation term. The selected-layout excess-risk term is $2\Delta_{\mathrm{unif}} + \epsilon_{\mathrm{opt}}$, so the table is a validation-burden diagnostic rather than an empirical dominance claim by itself.

Table A.9
Theory-to-evidence bridge for TRACE-BiOpt. The table does not claim that empirical diagnostics prove the theorems. It shows that each formal statement is tied to a visible current-best evidence lane and a bounded reviewer-facing interpretation.

Label	Formal statement	Paper-visible evidence	Current-best readout	Reviewer-facing interpretation
T1	MAP closed form and stability	MAP stability posture table	6/9 rows lower condition on all 10 splits; max TRACE-BiOpt condition number 7026.	Current-best gains do not require numerically pathological GLS/MAP solves.
T2	Posterior trace as Bayes squared risk	Posterior-risk posture probes	4/4 matched probes show lower posterior trace with lower held-out MAE.	The posterior certificate is visibly aligned with downstream reconstruction quality in bounded matched-route probes.
T3	Uniform generalization over all size-k layouts	Generalization-burden table	$B^{-1}\Delta_{\mathrm{unif}}$ spans 0.263 (PeMS7_228 10%) to 0.770 (PeMS7_1026 30%).	The theory explains why the largest-network rows need the strongest calibration/search support under the same validation window.
T4	Exchange certificate over the searched neighborhood	Exchange-certificate table + bounded exact benchmark	Certificate scopes cover budget-limited certificate, complete one-exchange certificate, mixed searched certificate, searched active-set certificate; bounded exact benchmark exact-hits 27/27.	The solver certificate is explicit about searched-neighborhood scope and is cross-checked on audited small subnetworks.
T5	Continuous-relaxation consistency	Initializer-posture table	Initializer families in evidence: objective_forward, posterior_greedy_warm_start; stage modes: exchange_only_warm_start, forward_then_exchange.	Relaxation remains a warm-start story inside one objective, not a second evidence-backed method hiding beside TRACE-BiOpt.
T6	CVaR tail-risk epigraph and interpretation	Tail-risk and weight-sensitivity posture tables	Current-best CVaR share stays between 0.26% and 1.46%, yet removing or mis-weighting it worsens held-out MAE in every matched probe.	The tail term is small but operative: a coherent risk penalty, not decorative notation.

I. Challenger envelope curves

Figure A.2 compresses the same current-best evidence into a strongest-challenger envelope: for each sensor budget, the envelope traces the strongest pre-registered non-BiOpt comparator, which may differ across budgets. TRACE-BiOpt stays below the row-specific strongest pre-registered non-BiOpt comparator at every budget on all three networks.

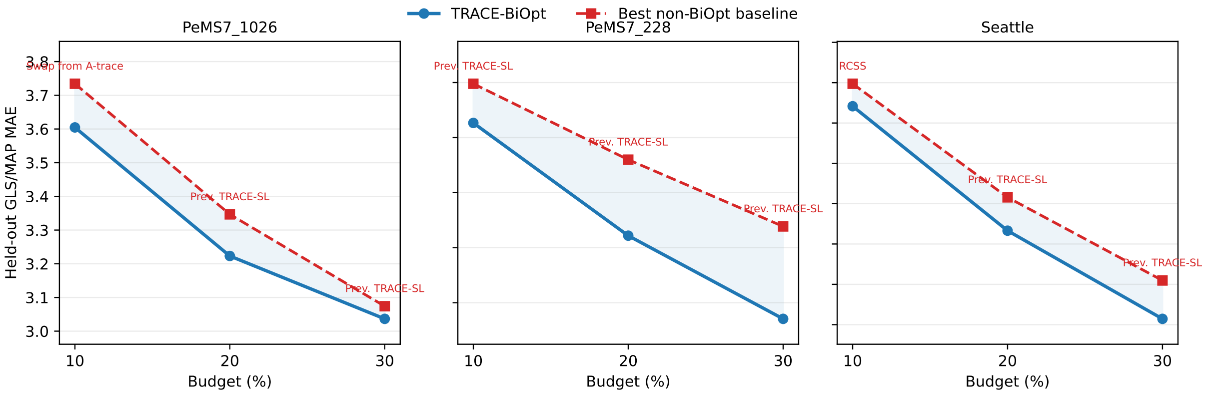

Figure A.2: Strongest-challenger envelope across sensor budgets. Each panel plots TRACE-BiOpt against the row-specific strongest pre-registered non-BiOpt baseline from the current-best evidence chain. The strongest challenger may differ across budgets. TRACE-BiOpt remains the lower-MAE curve on every dataset-budget row.

J. Mechanism diagnostic figures

Table A.10
PeMS7_228 component ablation across budgets. Values are ten-split held-out GLS/MAP MAE; lower is better.

Layout component	10%	20%	30%
Random mean	3.833	3.575	3.404
Best random by validation	3.713	3.450	3.247
Greedy A-trace	3.686	3.465	3.351
RCSS selected	3.605	3.393	3.183
TRACE-SL	**3.590**	**3.355**	**3.084**
Multistart swap	3.595	3.397	3.204

Table A.11
Reviewer-facing TRACE-BiOpt route-ablation slices. Unlike the legacy TRACE-SL component table, these rows stay inside the current TRACE-BiOpt objective family and ask which route ingredient is being removed or weakened on the rows that mattered most to the final claim chain.

Case	Intervention family	Current route	Ablated route	What changes	Current MAE	Ablated MAE	Δ MAE	Reading
PeMS7_228 10%	Risk calibration + lighter certificate weights	Calibrated low-cert	Stage15 certified	train+val risk, (0.50,0.01,0.00), full-search 192 vs val risk, (2.00,0.05,0.01), Stage15 active-set	3.248	3.377	0.129	The mixed low-budget row is resolved inside the same TRACE-BiOpt family by calibrating upper-level risk, lowering cert weights, and completing full exchange.
PeMS7_228 10%	Certificate family removal	Calibrated low-cert	Zero-weight probe	same split, but (0,0,0) under strong-search 96	3.248	3.417	0.169	The promoted row is not explained by dropping certificates entirely; the zero-weight strong-search route is still worse.
Seattle 20%	Certificate family removal	Seattle diagnostic certified	Zero-weight probe	remove posterior/CVaR/spatial terms under strong-search 96	2.674	2.710	0.036	Stable Seattle diagnostic slices still benefit from the certificate-weighted objective even when search is already strong.
Seattle 30%	Certificate family removal	Seattle diagnostic certified	Zero-weight probe	remove posterior/CVaR/spatial terms under strong-search 96	2.488	2.520	0.032	The stable high-budget Seattle diagnostic slice keeps the same sign: strong search alone does not recover the certificate-weighted route.
PeMS7_1026 30%	Risk calibration + stronger search tail	Calibrated rerun	Stage15 default	train+val risk, low-cert rerun, stronger active-set exchange vs original Stage15 route	3.037	3.071	0.035	The large-network weak row tightens inside the same objective family rather than by switching to another baseline family.

Δ MAE is the held-out penalty of the ablated route relative to the reference TRACE-BiOpt route listed in the table, so positive values favor that reference route. The Seattle rows are bounded same-split diagnostic slices built from the matched strong-search probes already used in the certificate-removal and weight-sensitivity tables; they do not replace the ten-seed current-best evidence chain. The PeMS7_228 and PeMS7_1026 rows use the completed calibrated reruns that now feed the current-best evidence chain. This is bounded route-ablation evidence, not a new dominance table.

Table A.12
Mechanism diagnostics for TRACE-BiOpt from the current best evidence chain.

Diagnostic	Evidence	Interpretation
Strongest-comparator gate	TRACE-BiOpt beats the row-specific best non-BiOpt baseline in 9/9 rows; 21 baseline rows per regime.	The main table is not a comparison against a fixed weak baseline.
Paired-test strength	9/9 rows have paired t-test $p<0.05$; 9/9 rows win all ten paired splits.	Most regimes show split-paired evidence beyond mean dominance.
Weak-row resolution path	PeMS7_1026 30%: delta -0.0375 MAE, 10/10 paired wins, paired t p=0.0000; 8/9 current-best rows are already promoted from calibrated reruns.	Weak rows are resolved inside the same TRACE-BiOpt objective family and then promoted into the audited main evidence chain.
Comparator hardness	rcss_selected in 1/9 rows, swap_from_greedy_a_trace in 1/9 rows, validation_swap_selected in 7/9 rows.	The strongest comparator often comes from the prior TRACE-SL or RCSS family, not only simple heuristics.
Certificate alignment	posterior trace Spearman rho=0.849; condition number rho=0.865; information logdet rho=-0.778; n=30.	Posterior/certificate quantities track held-out MAE under the GLS/MAP protocol.

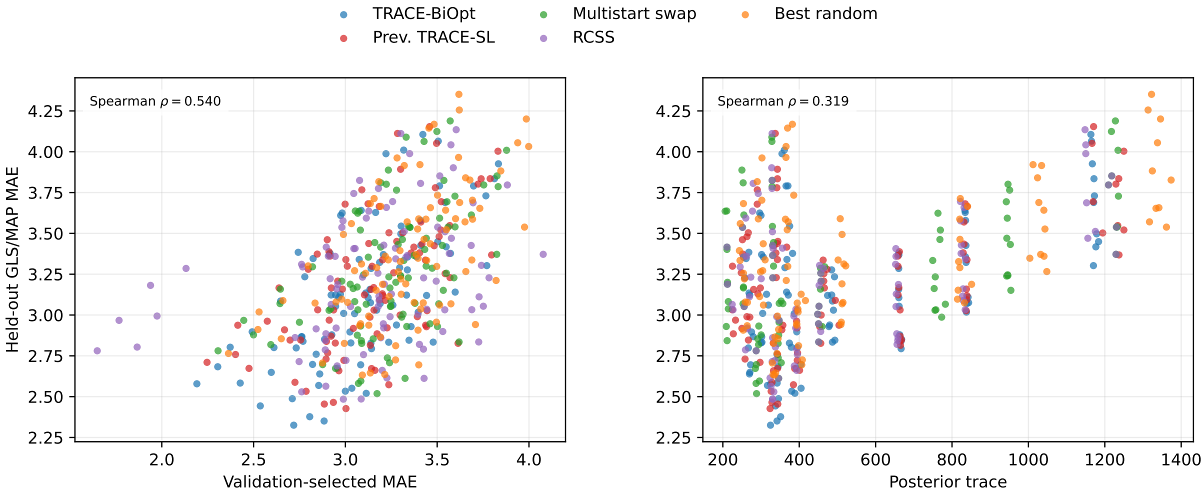

Figure A.3: Mechanism-alignment diagnostics on the 450 GLS/MAP selected-layout rows formed by three networks, three budgets, ten seeds, and five layout families. Left: validation-selected MAE versus held-out MAE. Right: posterior trace versus held-out MAE on the same selected layouts.

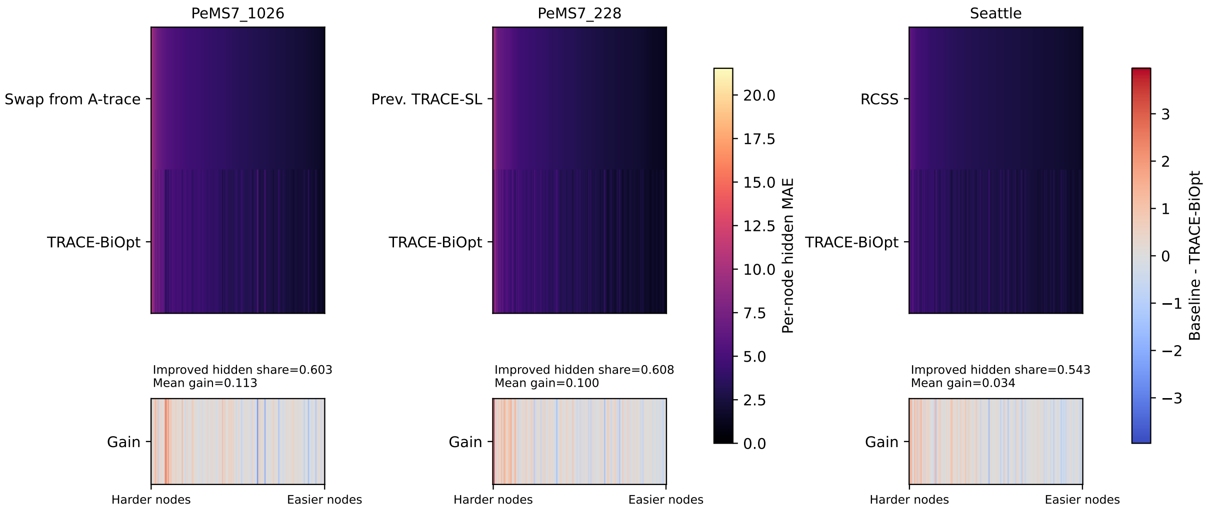

Figure A.4: Representative low-budget common-hidden error heatmaps on the seed-25 current-best rows. Within each dataset, columns are the nodes that remain hidden under both layouts, ordered from higher to lower strongest-baseline hidden-node MAE. The top heatmap compares node-wise hidden MAE for the strongest baseline and TRACE-BiOpt; the bottom strip shows baseline minus TRACE-BiOpt, so positive values favor TRACE-BiOpt.

K. Layout visualization

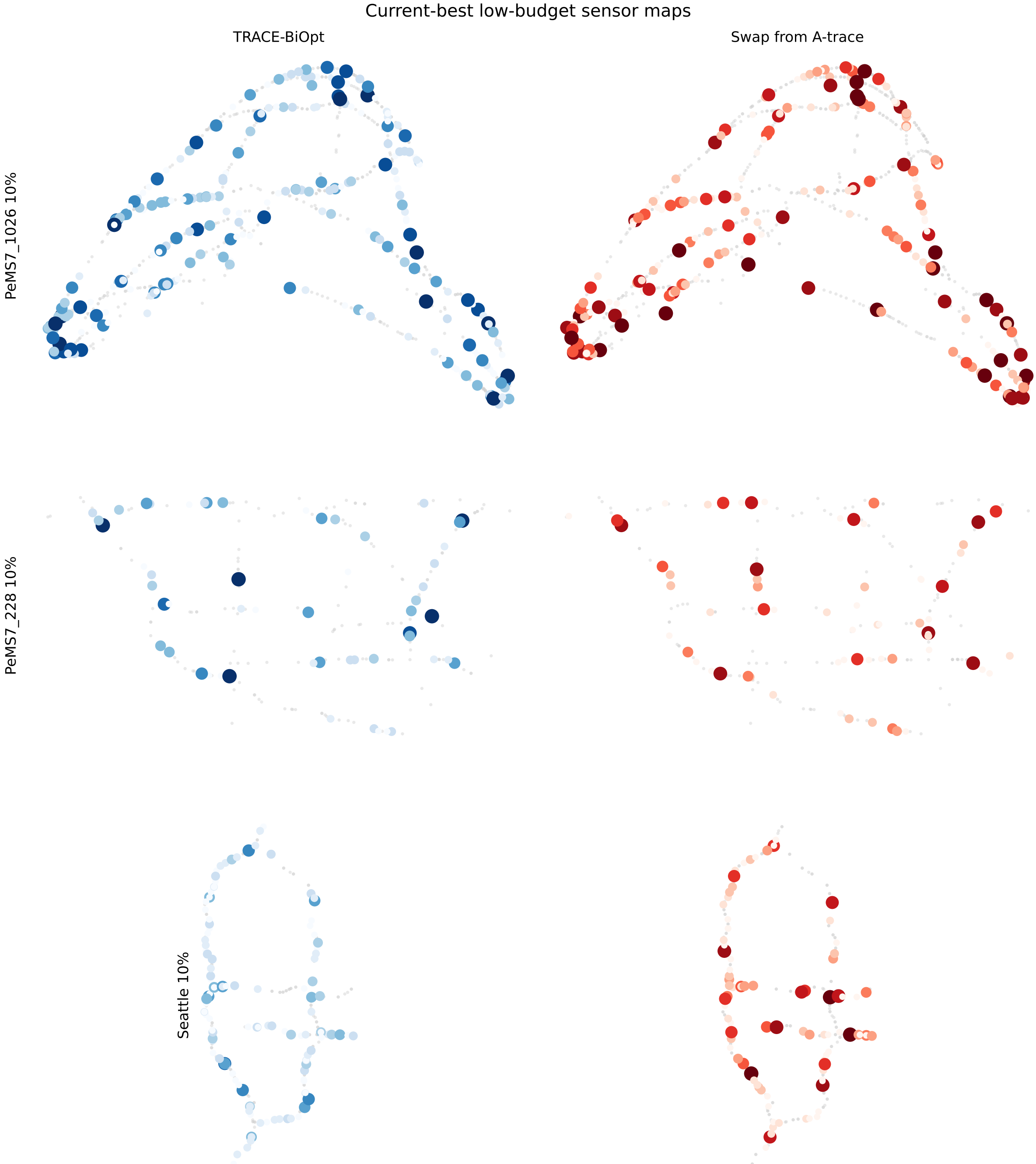

Figure A.5: Current-best low-budget sensor maps. PeMS7_228 and Seattle use released geographic sensor coordinates; PeMS7_1026 uses a distance-derived network embedding because benchmark station coordinates are not released with the public artifact. Darker and larger markers indicate nodes selected in more of the ten split-specific layouts. Dotted edges in each panel are visual aids for spatial readability and are not intended to represent the complete road-topology.

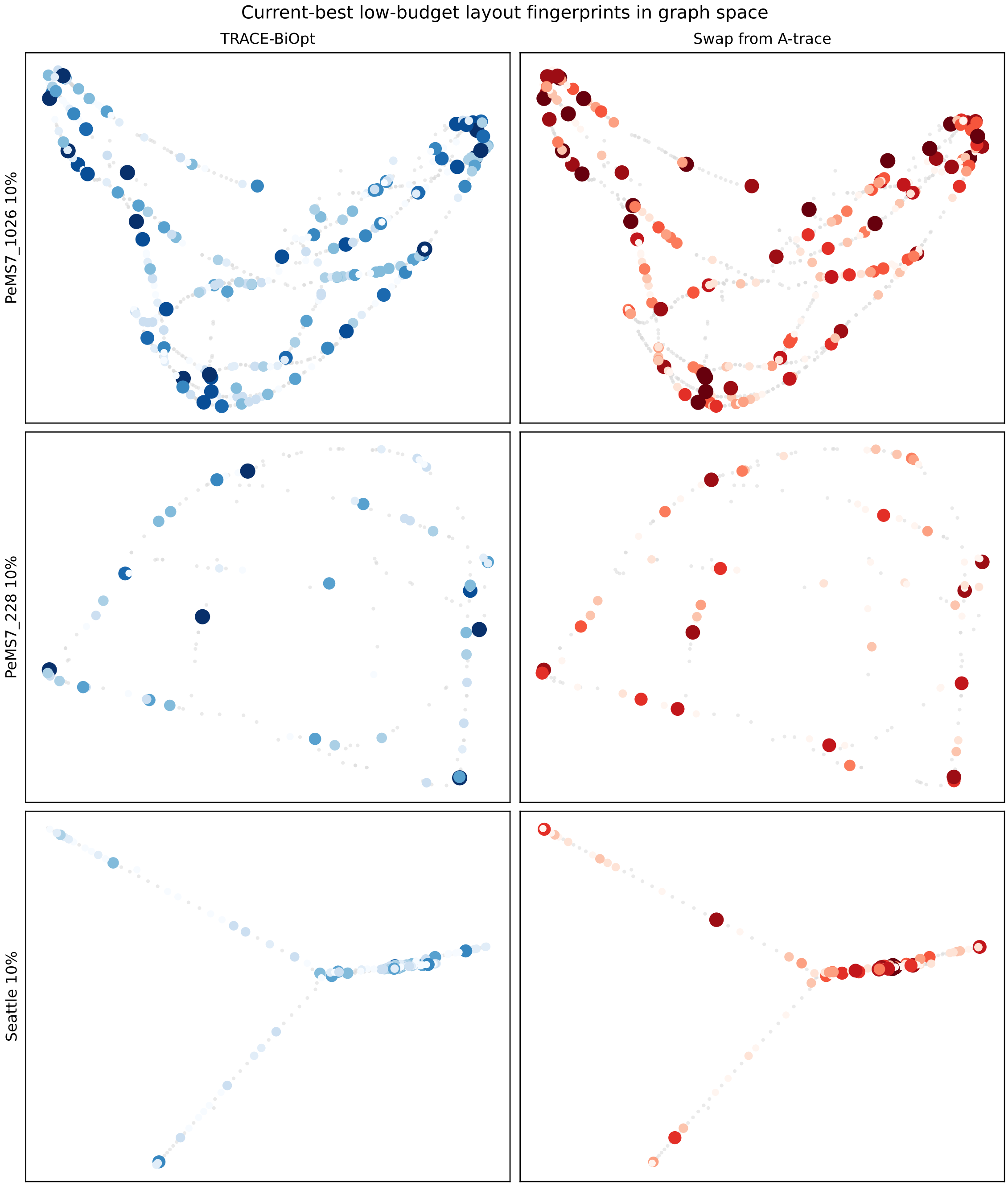

Figure A.6: Current-best low-budget layout fingerprints in graph space. Each row shows the 10% budget regime for one dataset; node positions come from a deterministic spectral embedding of the network similarity graph, and darker/larger markers indicate sensors selected in more of the ten split-specific layouts.

L. Solver diagnostics

Table A.13
TRACE-BiOpt optimization diagnostics from Stage15 histories, including searched one-exchange coverage and stop certificates.

Dataset	Budget	Exchanges	Search cov.	Stop cert.	Full drop	Exchange drop	Mono.
PeMS7_1026	10%	3.0 [3,3]	0.2%	0/10	0.9%	0.89%	10/10
PeMS7_1026	20%	3.0 [3,3]	0.1%	0/10	0.8%	0.81%	10/10
PeMS7_1026	30%	3.0 [3,3]	0.1%	0/10	0.8%	0.82%	10/10
PeMS7_228	10%	1.7 [0,3]	4.1%	6/4	57.7%	0.33%	10/10
PeMS7_228	20%	1.3 [0,3]	2.3%	7/3	64.8%	0.33%	10/10
PeMS7_228	30%	1.0 [0,3]	1.8%	9/1	68.7%	0.06%	10/10
Seattle	10%	5.4 [0,8]	13.7%	9/1	60.0%	0.10%	10/10
Seattle	20%	5.1 [2,8]	7.6%	7/3	67.7%	0.39%	10/10
Seattle	30%	4.1 [0,7]	5.8%	10/0	72.1%	0.00%	10/10

Each row aggregates ten Stage15 split-seed runs. Search cov. is the mean searched one-exchange neighborhood size as a percentage of the full $k(n-k)$ swap set under the declared deterministic add/remove pools. Stop cert. reports runs ending by no improving searched one-exchange move versus runs exhausting the declared exchange-iteration budget. Full drop is the mean percent reduction from the first recorded construction or warm-start objective to the final TRACE-BiOpt objective. Exchange drop is the mean percent reduction from the completed construction/warm-start layout to the final exchanged layout. Monotone counts runs whose accepted construction/exchange history is non-increasing in the single TRACE-BiOpt objective.

Table A.14
Current-best TRACE-BiOpt solver scale by dataset-budget row.

Dataset	Budget	Source	Init.	Exchanges	Search cov.	Peak RSS	Stop cert.
PeMS7_1026	10%	Stage16 calibrated rerun	posterior_greedy_warm_start	18.9 [16,20]	1.3%	1.39 [1.29,1.43] GB	4/6
PeMS7_1026	20%	Stage16 calibrated rerun	posterior_greedy_warm_start	20.0 [20,20]	0.8%	2.87 [2.78,2.92] GB	0/10
PeMS7_1026	30%	Stage16 calibrated rerun	posterior_greedy_warm_start	8.0 [8,8]	0.6%	1.50 [1.49,1.51] GB	0/10
PeMS7_228	10%	Stage16 calibrated rerun	objective_forward	7.1 [4,10]	100.0%	0.40 [0.33,0.52] GB	10/0
PeMS7_228	20%	Stage16 calibrated rerun	objective_forward	11.3 [9,12]	100.0%	0.72 [0.66,0.73] GB	4/6
PeMS7_228	30%	Stage16 calibrated rerun	objective_forward	11.9 [11,12]	100.0%	1.56 [1.49,1.57] GB	1/9
Seattle	10%	Stage15 main evidence	objective_forward	5.4 [0,8]	13.7%	0.27 [0.23,0.29] GB	9/1
Seattle	20%	Stage16 calibrated rerun	objective_forward	3.7 [0,7]	7.6%	0.37 [0.30,0.52] GB	10/0
Seattle	30%	Stage16 calibrated rerun	objective_forward	3.0 [0,7]	5.8%	0.45 [0.37,0.52] GB	10/0

Source identifies whether the current-best row is still backed by Stage15 main evidence or by a promoted Stage16 calibrated rerun. Init. is the dominant deterministic initializer recorded across the ten split-specific runs. Search cov. is the mean searched one-exchange neighborhood size as a percentage of the full $k(n-k)$ swap set under the declared add/remove pools, with zero-valued pools interpreted as complete enumeration. Peak RSS is the mean, minimum, and maximum peak resident memory observed during TRACE-BiOpt search events for that row. Stop cert. reports runs ending by no improving searched exchange versus runs exhausting the declared exchange-iteration budget.

Table A.15
Current-best TRACE-BiOpt exchange-gap certificate posture by dataset-budget row.

Dataset	Budget	Search cov.	Stop cert.	Certificate scope
PeMS7_1026	10%	1.3%	4/6	mixed searched certificate
PeMS7_1026	20%	0.8%	0/10	budget-limited certificate
PeMS7_1026	30%	0.6%	0/10	budget-limited certificate
PeMS7_228	10%	100.0%	10/0	complete one-exchange certificate
PeMS7_228	20%	100.0%	4/6	mixed searched certificate
PeMS7_228	30%	100.0%	1/9	mixed searched certificate
Seattle	10%	13.7%	9/1	mixed searched certificate
Seattle	20%	7.6%	10/0	searched active-set certificate
Seattle	30%	5.8%	10/0	searched active-set certificate

Search cov. is the mean searched one-exchange neighborhood as a percentage of the full $k(n-k)$ swap set. Stop cert. reports runs ending by no improving searched exchange versus runs exhausting the declared exchange-iteration budget. A complete one-exchange certificate means $G_1(\hat{S}) = 0$ up to tolerance because the full neighborhood was searched; a searched active-set or mixed certificate means only $G_1^{\text{search}}(\hat{S}) = 0$ on the evaluated active set; a budget-limited row does not carry a zero-gap certificate.

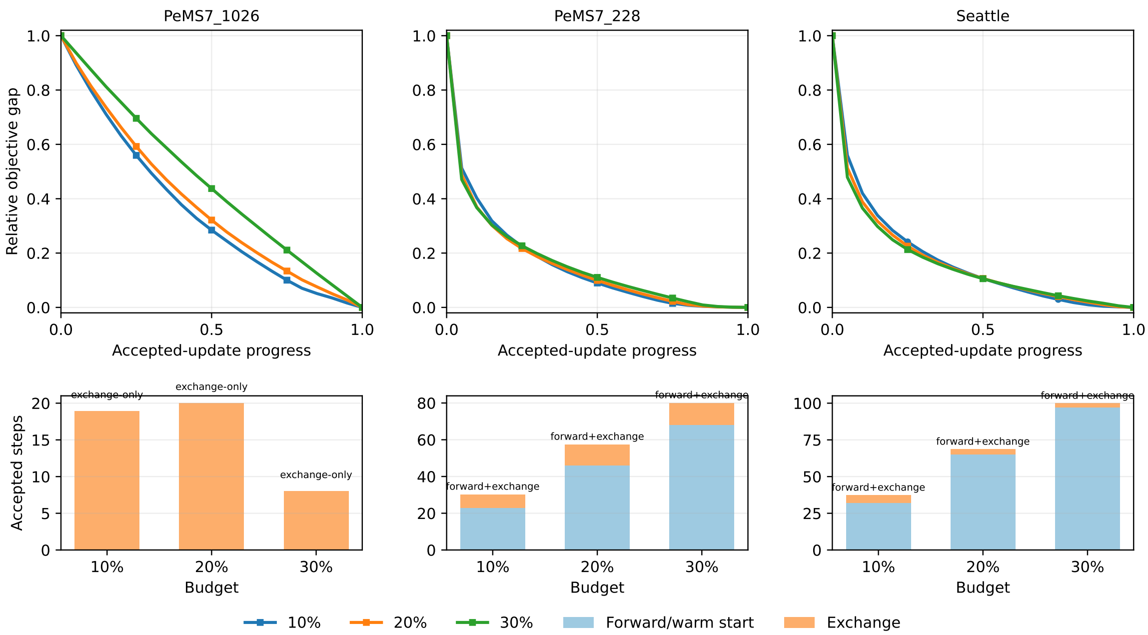

Figure A.7: Current-best TRACE-BiOpt objective descent and accepted-step mix. The top row shows the mean relative objective gap remaining over normalized accepted-update progress; the bottom row stacks the mean accepted forward/warm-start and exchange steps for each budget.

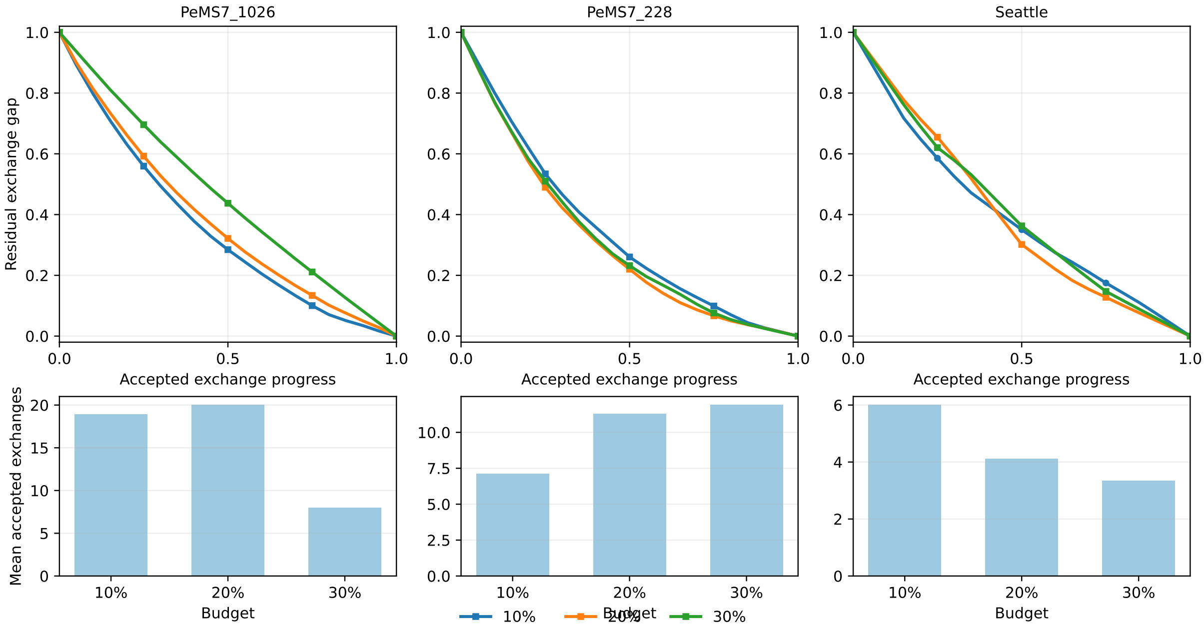

Figure A.8: Current-best TRACE-BiOpt residual exchange-gap profiles. The top row shows the mean fraction of eventual exchange-phase improvement still remaining after the accepted exchange phase begins; the bottom row reports the mean number of accepted exchange steps, with '0-ex' annotations giving how many of the ten split-specific runs accepted no exchange at all beyond the forward or warm-start layout.

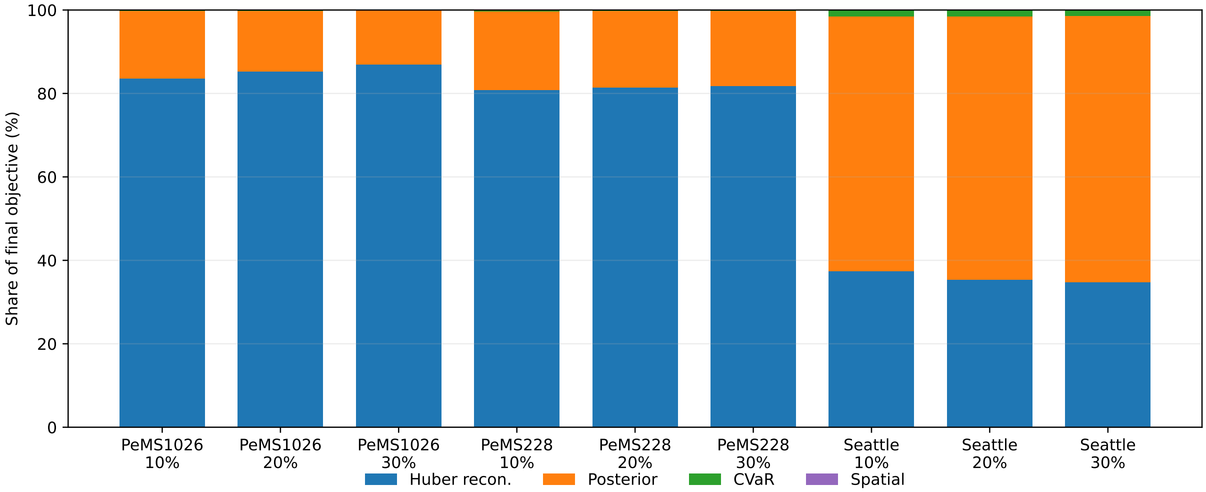

Figure A.9: Current-best TRACE-BiOpt final-objective composition. Each bar shows the mean weighted share of the accepted final objective coming from hidden Huber reconstruction loss, posterior trace, scenario-CVaR trace, and spatial regularization over the ten split-specific runs of that row.

Table A.16
PeMS7_1026 30% weak-row enhanced-search evidence over the original Stage15 ten split seeds. The enhanced TRACE-BiOpt rerun keeps the same objective and increases only deterministic exchange-search pools/iterations. Baseline values are the seed-matched pre-registered non-BiOpt rows from the Stage15 evidence path. This row-level diagnostic strengthens the weakest regime but does not replace the full nine-regime Stage15 main table.

Seed	Default Δ	Enhanced Δ	MAE gain	Best baseline	Enhanced beats?	
25	-0.0174	-0.0473	0.0298	validation_swap_selected	yes	
26	-0.0110	-0.0410	0.0300	validation_swap_selected	yes	
27	0.0129	-0.0097	0.0226	validation_swap_selected	yes	
28	-0.0185	-0.0432	0.0246	greedy_a_trace	yes	
29	0.0225	-0.0176	0.0401	validation_swap_selected	yes	
30	-0.0082	-0.0270	0.0187	validation_swap_selected	yes	Enhanced-search
31	-0.0097	-0.0487	0.0390	validation_swap_selected	yes	
32	0.0085	-0.0215	0.0301	validation_swap_selected	yes	
33	0.0088	-0.0358	0.0446	rcss_selected	yes	
34	-0.0135	-0.0457	0.0321	validation_swap_selected	yes	
Mean	-0.0026	-0.0338	0.0312	–	10/10	

paired t-test versus zero delta: $p = 1.5e-05$; Wilcoxon signed-rank $p = 0.0010$; mean MAE-gain t-test $p = 3.6e-07$. Negative Δ means TRACE-BiOpt has lower held-out GLS/MAP MAE than the seed-matched best pre-registered non-BiOpt baseline.

Table A.17
PeMS7_228 10% low-budget calibration-risk diagnostic over the original Stage15 ten split seeds. The calibrated TRACE-BiOpt rerun keeps a single reconstruction-aware objective, estimates upper-level reconstruction risk on train+validation days, lowers certificate weights, and uses complete one-exchange search. Baseline values are the seed-matched pre-registered non-BiOpt rows from the Stage15 evidence path. This row-level diagnostic addresses the second mixed paired-evidence row but does not replace the full nine-regime Stage15 main table.

Seed	Default Δ	Calibrated Δ	MAE gain	Best baseline	Calibrated beats?
25	-0.0207	-0.1497	0.1291	swap_from_scenario_average	yes
26	0.0283	-0.0514	0.0797	scenario_cvar_a_trace	yes
27	-0.0095	-0.0483	0.0388	multistart_swap_by_validation	yes
28	-0.0711	-0.1375	0.0664	best_random_by_validation	yes
29	0.0937	-0.0537	0.1474	validation_swap_selected	yes
30	-0.0673	-0.1385	0.0712	rcss_selected	yes
31	0.1175	-0.0455	0.1630	validation_swap_selected	yes
32	-0.1534	-0.1078	-0.0456	robust_coverage_cvar	yes
33	-0.0946	-0.1871	0.0925	scenario_average_a_trace	yes
34	0.0154	-0.1027	0.1181	rcss_selected	yes
Mean	-0.0162	-0.1022	0.0860	–	10/10

Calibrated-risk paired t-test versus zero delta: $p = 6.4e-05$; Wilcoxon signed-rank $p = 0.0010$; mean MAE-gain t-test $p = 7.4e-04$. Negative Δ means TRACE-BiOpt has lower held-out GLS/MAP MAE than the seed-matched best pre-registered non-BiOpt baseline.

Table A.18
PeMS7_1026 30% Stage16 calibrated-risk diagnostic over the original Stage15 ten split seeds. The rerun uses the same train+validation risk-source refinement as Table A.17 with scalable active-set exchange search for the 1026-node network. Baseline values are seed-matched pre-registered non-BiOpt rows from the Stage15 evidence path. This row-level diagnostic strengthens the weakest main-table regime but does not replace the full nine-regime Stage15 main table.

Seed	Default Δ	Calibrated Δ	MAE gain	Best baseline	Calibrated beats?
25	-0.0174	-0.0608	0.0434	validation_swap_selected	yes
26	-0.0110	-0.0487	0.0377	validation_swap_selected	yes
27	0.0129	-0.0045	0.0174	validation_swap_selected	yes
28	-0.0185	-0.0517	0.0332	greedy_a_trace	yes
29	0.0225	-0.0227	0.0452	validation_swap_selected	yes
30	-0.0082	-0.0245	0.0163	validation_swap_selected	yes
31	-0.0097	-0.0499	0.0401	validation_swap_selected	yes
32	0.0085	-0.0225	0.0311	validation_swap_selected	yes
33	0.0088	-0.0407	0.0495	rcss_selected	yes
34	-0.0135	-0.0456	0.0320	validation_swap_selected	yes
Mean	-0.0026	-0.0372	0.0346	–	10/10

Calibrated-risk paired t-test versus zero delta: $p = 4.6e-05$; Wilcoxon signed-rank $p = 0.0010$; mean MAE-gain t-test $p = 2.0e-06$. Negative Δ means TRACE-BiOpt has lower held-out GLS/MAP MAE than the seed-matched best pre-registered non-BiOpt baseline.

Table A.19
Reviewer-facing regime lessons from the current-best TRACE-BiOpt evidence chain. The table translates row-level strongest-baseline, replacement-route, and optimization-certificate evidence into transport-design interpretation.

Regime	Strongest non-BiOpt family	Evidence lane	Search signal	Main lesson
PeMS7_1026 10%	A-trace swap family	Stage16 calibrated rerun	search-budget sensitive	Direct A-trace local search is insufficient once hidden-state risk, tail risk, and redundancy are optimized jointly.
PeMS7_1026 20%	prior TRACE-SL swap family	Stage16 calibrated rerun	search-budget sensitive	The regime benefits from more aggressive searched one-exchange refinement because the declared budget often binds before local stability.
PeMS7_1026 30%	prior TRACE-SL swap family	Stage16 calibrated rerun	search-budget sensitive	Large networks need both calibrated risk estimation and more search budget to convert a weak margin into a stable paired win.
PeMS7_228 10%	prior TRACE-SL swap family	Stage16 calibrated rerun	searched-neighborhood stable	Thin validation windows can mis-rank low-budget layouts; calibrating reconstruction risk on train+validation resolves the row.
PeMS7_228 20%	prior TRACE-SL swap family	Stage16 calibrated rerun	searched-neighborhood stable	The dominant gain is robust direct optimization of reconstruction risk over the prior TRACE-SL swap family under the same evaluation protocol.
PeMS7_228 30%	prior TRACE-SL swap family	Stage16 calibrated rerun	searched-neighborhood stable	The dominant gain is robust direct optimization of reconstruction risk over the prior TRACE-SL swap family under the same evaluation protocol.
Seattle 10%	RCSS selector	Stage15 main evidence	searched-neighborhood stable	TRACE-BiOpt beats a certificate-ranked selector directly, so the gain is not only from candidate-pool curation.
Seattle 20%	prior TRACE-SL swap family	Stage16 calibrated rerun	searched-neighborhood stable	The dominant gain is robust direct optimization of reconstruction risk over the prior TRACE-SL swap family under the same evaluation protocol.
Seattle 30%	prior TRACE-SL swap family	Stage16 calibrated rerun	searched-neighborhood stable	The dominant gain is robust direct optimization of reconstruction risk over the prior TRACE-SL swap family under the same evaluation protocol.

M. Budget phasing and bounded exact benchmark

Table A.20
Current-best TRACE-BiOpt budget-phasing posture. Each row turns the 10%, 20%, and 30% current-best curves into a staged deployment readout rather than another dominance table.

Dataset	TRACE-BiOpt MAE path	TRACE-BiOpt marginal gain	Margin path vs strongest baseline	Deployment readout
PeMS7_1026	3.605 -> 3.223 -> 3.037	10->20: 0.382; 20->30: 0.187; first-step share: 67%	0.130 -> 0.124 -> 0.038	The first 10-point increment captures 67% of the total 10-30% TRACE-BiOpt gain, but the strongest-baseline margin still contracts at 30% (0.130 -> 0.124 -> 0.038), so extra budget still needs calibrated search scaling.
PeMS7_228	3.453 -> 3.044 -> 2.741	10->20: 0.410; 20->30: 0.302; first-step share: 58%	0.142 -> 0.276 -> 0.336	The first increment captures 58% of the total gain, and the strongest-baseline margin keeps widening (0.142 -> 0.276 -> 0.336), so added sensors keep paying in both absolute and relative recoverability.
Seattle	3.042 -> 2.733 -> 2.514	10->20: 0.309; 20->30: 0.219; first-step share: 59%	0.056 -> 0.083 -> 0.095	The first increment captures 59% of the total gain and the strongest-baseline margin widens monotonically (0.056 -> 0.083 -> 0.095), making staged rollout the cleanest on Seattle.

Table A.21
Bounded exact-subnetwork benchmark for TRACE-BiOpt. Each dataset uses three deterministic anchor-centered 16-node induced subnetworks. For each subnetwork, the paper reuses the row-wise 10%, 20%, and 30% TRACE-BiOpt route parameters and exhaustively enumerates every feasible k-subset.

Dataset	Exact benchmark setup	Exact hit count	Objective / MAE gap	Reviewer readout
PeMS7_1026	3 anchors, 16 nodes each; 10/20/30% -> 2/3/5 sensors; 120 / 560 / 4368 layouts per anchor.	9/9 exact hits	obj mean/max 0.0000/0.0000; MAE mean/max 0.0000/0.0000	Even the search-budget-sensitive PeMS7_1026 rows hit the exact optimum on bounded 16-node subnetworks, so the remaining solver caveat is about scale, not local route pathology.
PeMS7_228	3 anchors, 16 nodes each; 10/20/30% -> 2/3/5 sensors; 120 / 560 / 4368 layouts per anchor.	9/9 exact hits	obj mean/max 0.0000/0.0000; MAE mean/max 0.0000/0.0000	The calibrated PeMS7_228 route stays exact across all audited subnetworks, so the promoted low- and mid-budget rows are not relying on approximate local choices.
Seattle	3 anchors, 16 nodes each; 10/20/30% -> 2/3/5 sensors; 120 / 560 / 4368 layouts per anchor.	9/9 exact hits	obj mean/max 0.0000/0.0000; MAE mean/max 0.0000/0.0000	Seattle also exact-hits all audited subnetworks, but this does not upgrade the Seattle 10% main-row status: that row remains fail-closed on promotion evidence, not on small-subnetwork solver scope.